



EE5110/EE6110

**Special Topics in Automation and Control
Segment B**

From Frames to Events:
Theory and Applications of Event-based Vision

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Contacts

- You may contact me via

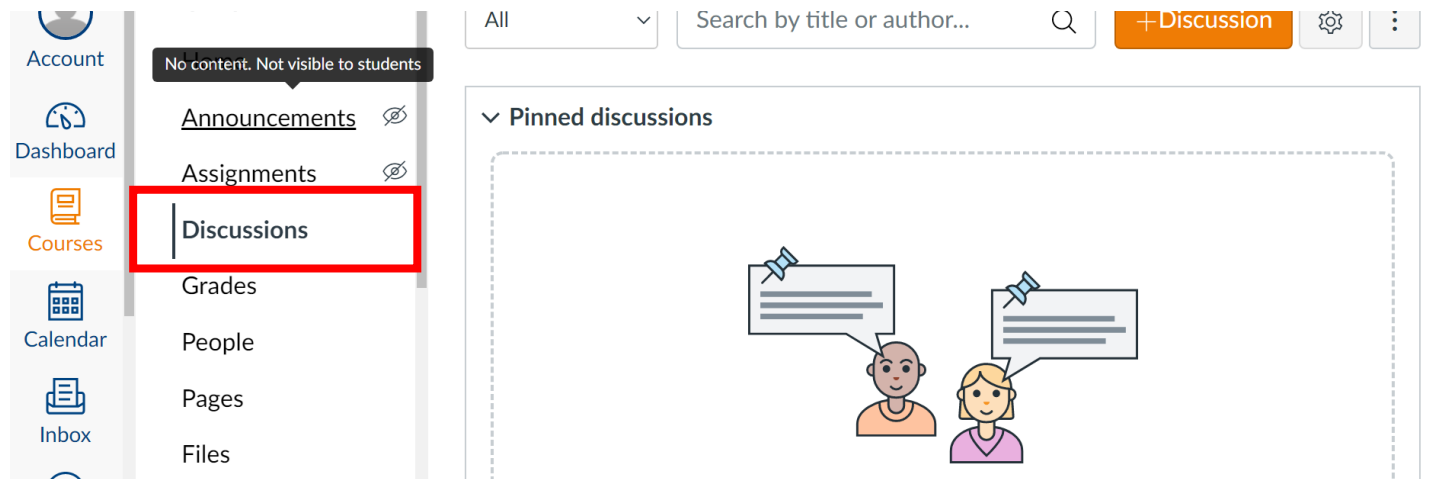
Phone **+65-89423081** (emergencies)

Email rui.j@nus.edu.sg (other issues)

- NUS Canvas

<https://canvas.nus.edu.sg/>

- Maintain communication after class
- A place for you to discuss anytime, anywhere, and to exchange information



Your feedback is valuable to us...

- Join the online student feedback exercise!
- Feedback report includes two parts:
 - Course report and teacher report
- **Course report**
 - Overall opinion, your expected grade, difficulty level
- **Teacher report**
 - Feedback to each lecturer

Question

The teacher has enhanced my thinking ability.

The teacher provided timely and useful feedback.

The teacher has increased my interest in the subject.

Raters	Student
Responded	17
Invited	26
Response Ratio	65%

Leave your (detailed) comments, not just scores.

Continuous Assessment (CA)

Important information:

- For CA, this is **group work**:
 - Form yourselves into groups of 1-5 students.
 - Each group should have one group leader.
 - Self-organize yourselves into groups **in CANVAS before the next lecture.**
- Submission for **CA: 4 parts**
 - 1) A presentation video (<15 mins), 2) presentation slides, 3) source code,
 - 4) **AI use and review report, if any AI tool used** (in .pdf format)
- Submit in one zip file with name: **your_group_leader_NUS_ID.zip**, e.g., A0123456X.zip.



[2410] 2024/2025 Semester 1

Home

Everyone

Segment B Group

Announcements

Self sign-up is enabled for these groups. (?)

Groups are limited to 5 members.

Modules

Continuous Assessment (CA)

- Contribution weight within a group

Final mark: Please attach the following table at the end of your slides. The table will be used to assess your contribution within the group. The final mark of each student FM will be based on your answers' mark M and be weighted according to your contribution weights:

$$FM_i = \frac{w_i}{\sum_{\forall i} w_i} \sum_{\forall i} M$$

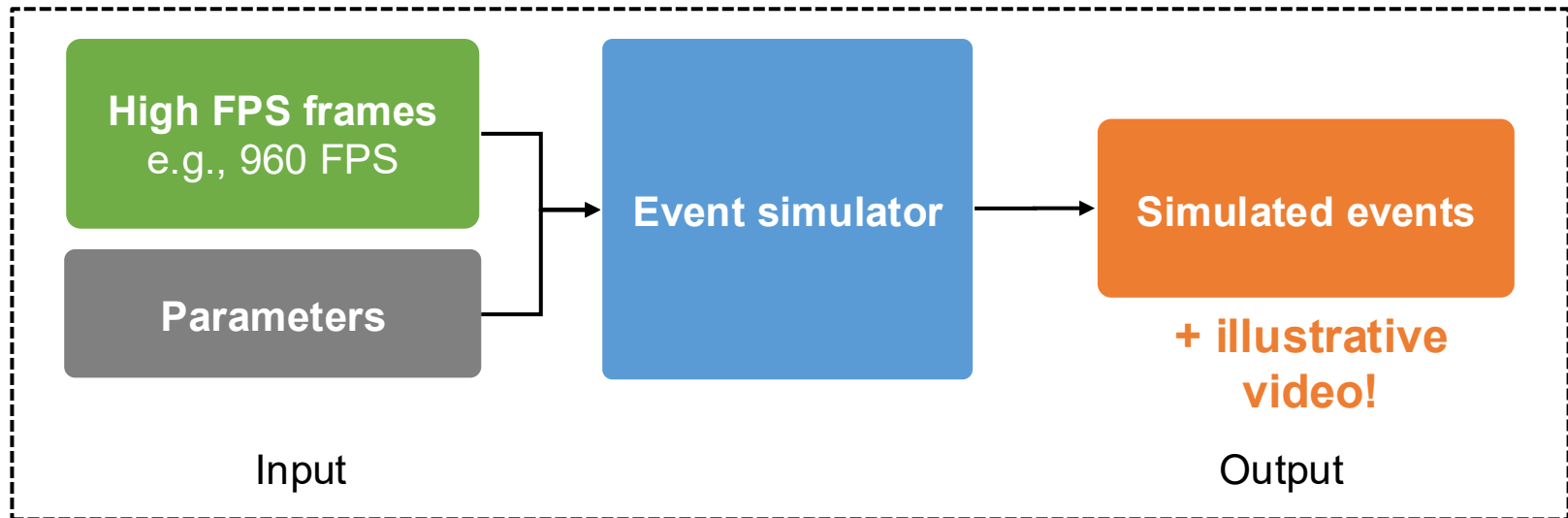
where i denotes the index of the student.

For equal contribution, simply put “1” for all contribution weights. Those answers without the table will be deemed as equal contribution.

No.	Name	Student number	Contribution weight
1 (team leader)			w_1
2			w_2
3			w_3

CA: Design an Event Camera Simulator

- **Aim:** To design a simulation framework to generate events, given high frame-rate images as inputs. A reference diagram is shown below:



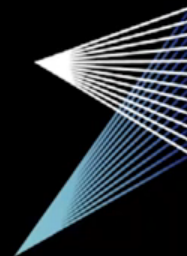
- The principle of event camera and event data simulation will be discussed later in the 1st lecture.

The Event-Camera Dataset: Event-based Data for Pose Estimation, Visual Odometry, and SLAM

Elias Mueggler, Henri Rebecq, Guillermo Gallego,
Tobi Delbruck and Davide Scaramuzza



**University of
Zurich**^{UZH}
Department of Informatics



**ROBOTICS &
PERCEPTION
GROUP**

rpg.ifi.uzh.ch

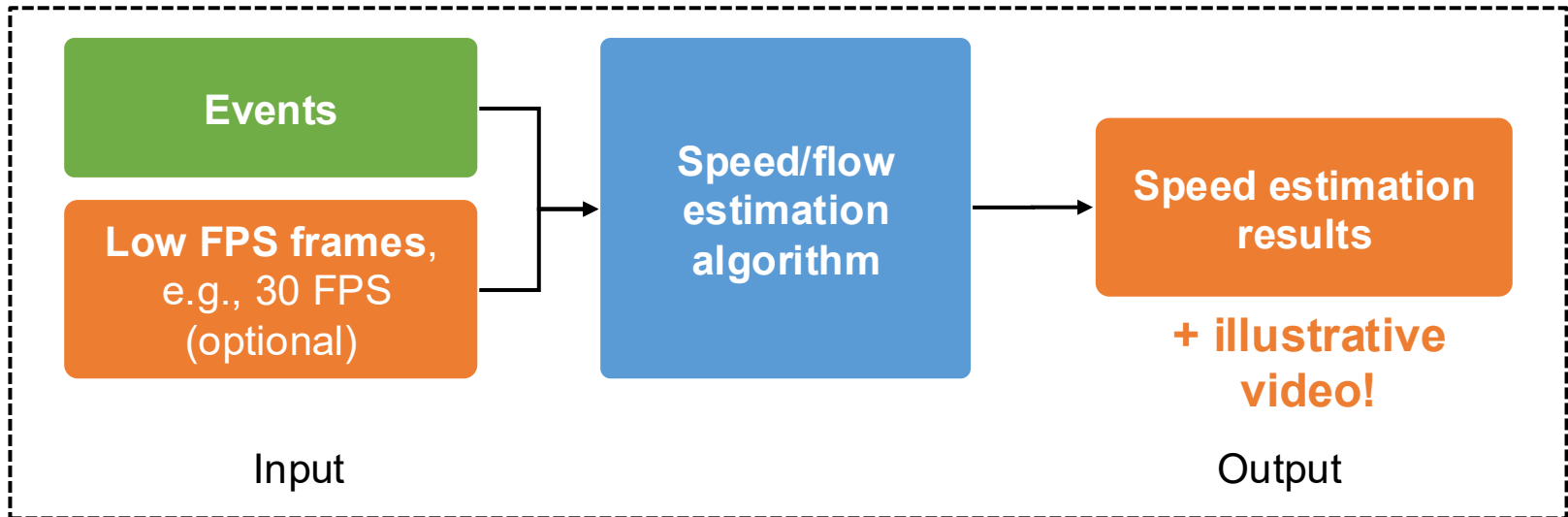
Research Project (RP)

Important information:

- For RP, this is an **individual project**.
- Submit in one zip file with name: **your_NUS_ID.zip**, e.g., A0123456X.zip.
- Submission for **RP: 4 parts**
 - 1) A report (≤ 6 pages), 2) demo video, 3) source code,
 - 4) **AI use and review report**, if any AI tool used (in .pdf format)
- Please always remain contactable during the semester, to arrange a 1-on-1 presentation:
 - 15~20 min presentation
 - 5 min live demo
 - 5~10 min Q&A

RP: Event-based Speed/Flow Estimation

- **Aim:** To design and implement an event-based speed/flow estimation framework. A reference diagram is shown below:



- Information hub for event-based vision:
[uzh-rpg/event-based_vision_resources \(github.com\)](https://github.com/uzh-rpg/event-based_vision_resources)
- More information/Q&A session in the 2nd lecture, after introduction of all required backgrounds/knowledge.

Secrets of Event-Based Optical Flow

Shintaro Shiba, Yoshimitsu Aoki, Guillermo Gallego

https://github.com/tub-rip/event_based_optical_flow



Continuous Assessment (CA) and Research Project (RP)

Important information:

- Written report / presentation should contain at least the following:
 1. An abstract that briefly summarizes what you have done and what the results are,
 2. a brief introduction to the project and the problem that you are solving,
 3. derivation of the model ("problem formulation"),
 4. description of the framework and algorithm(s) used
 5. the results and discussion,
 6. conclusions and/or a summary,
 7. references (if applicable).

Continuous Assessment (CA) and Research Project (RP)

- **AI Policy for this Segment**
 - AI tools are optional. If you use any AI tool, you must submit a concise **AI Use and Review Report** documenting its use and your review of the AI-generated content. **Otherwise, your submission will be regarded as entirely your own work without AI assistance.**
- The **AI Use and Review Report** should include:
 1. AI tool used
 2. Purpose of using AI
 3. 2-3 representative prompts (Full chat history is **not required**)
 4. Summary of useful AI feedback
 5. Changes made after AI feedback
 6. One AI suggestion not accepted, with reason
 7. Brief reflection on the reliability/limitations of AI

Continuous Assessment (CA) and Research Project (RP)

- **Be responsible to your submission**

- You are fully responsible for the correctness of the submitted code and outputs.
- You must understand and be able to explain your submitted work.
- You may use any accessible AI tool, subject to university and course guidelines: <https://ctltnus.edu.sg/ai-community-of-practice/policies/>



Continuous Assessment (CA) and Research Project (RP)

- Project end time and date: All results must be submitted by
CA - 11:59pm, Fri, 18 Sep 2026 (Week 6)
RP - 11:59pm, Fri, 6 Nov 2026 (Week 12)
 to CANVAS.
- RP presentation: in Week 13 / reading week

ACADEMIC CALENDAR AY2026/2027

				SEMESTER 1		
Regular Semester		Mini Semester		Week	Dates	
Semester: 18 weeks	Orientation			0	Mon, 3 Aug 2026	~ Sat, 8 Aug 2026
	Instructional Period (6 weeks)	Mini Sem 1A: 8 weeks	Instructional Period (6 weeks)	1	Mon, 10 Aug 2026	~ Fri, 14 Aug 2026
				2	Mon, 17 Aug 2026	~ Fri, 21 Aug 2026
				3	Mon, 24 Aug 2026	~ Fri, 28 Aug 2026
				4	Mon, 31 Aug 2026	~ Fri, 4 Sep 2026
				5	Mon, 7 Sep 2026	~ Fri, 11 Sep 2026
				6	Mon, 14 Sep 2026	~ Fri, 18 Sep 2026
	Recess				Sat, 19 Sep 2026	~ Sun, 27 Sep 2026
	Reading					
	Examination	7	Mon, 28 Sep 2026	~ Sat, 3 Oct 2026		

Course Structure

- **Lecture 1 – Event-based Vision System**
 - Conventional machine vision systems
 - Event-based sensors and cameras
 - Event camera calibration
 - Event camera autofocus
 - Event data simulation and dataset
 - Event data representation and compression

Course Structure

- **Lecture 2 – Algorithms and Applications**

- Detection and tracking
- Optical flow estimation
- Pose estimation and SLAM
- Image reconstruction
- Motion segmentation
- Motion recognition

Summary for Segment B

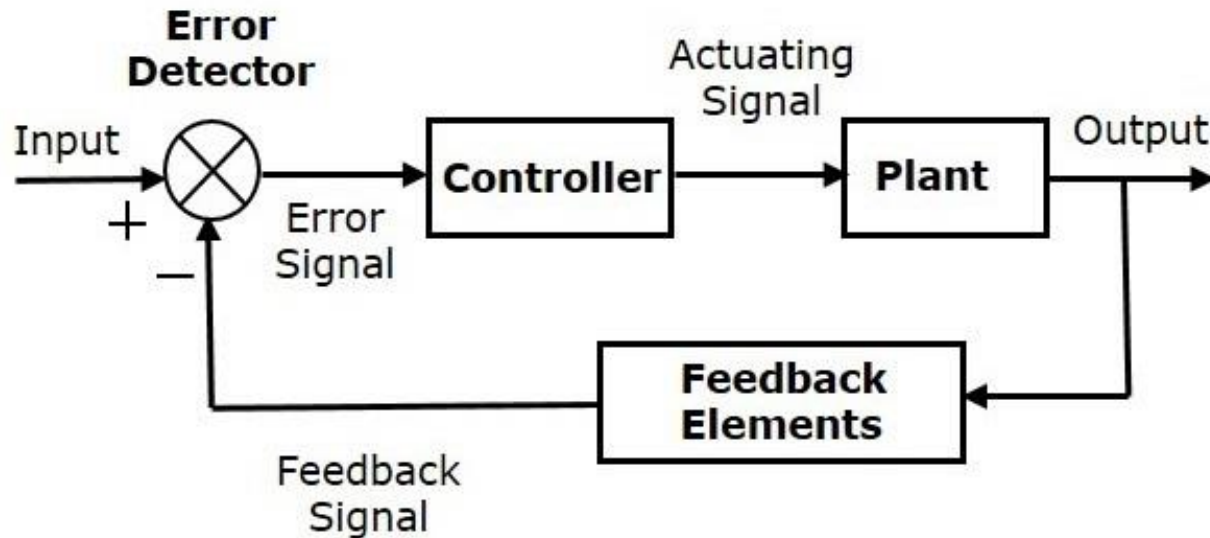
- An introductory course for event-based vision
- Seminar-like course with a wide range of topics
- Many papers, videos and applications to be given
- (hopefully, for Ph.D. students) Bringing ideas to research projects

Learning Objectives

- In this lecture, you are expected to:
 1. Describe and explain the basic concepts in both conventional and event-based machine vision systems;
 2. Understand the principle, pros and cons, and current status of event cameras;
 3. Understand the common procedure of event-based sensing, including data representation and data compression;
 4. Understand the mechanism and principles of event data simulation.

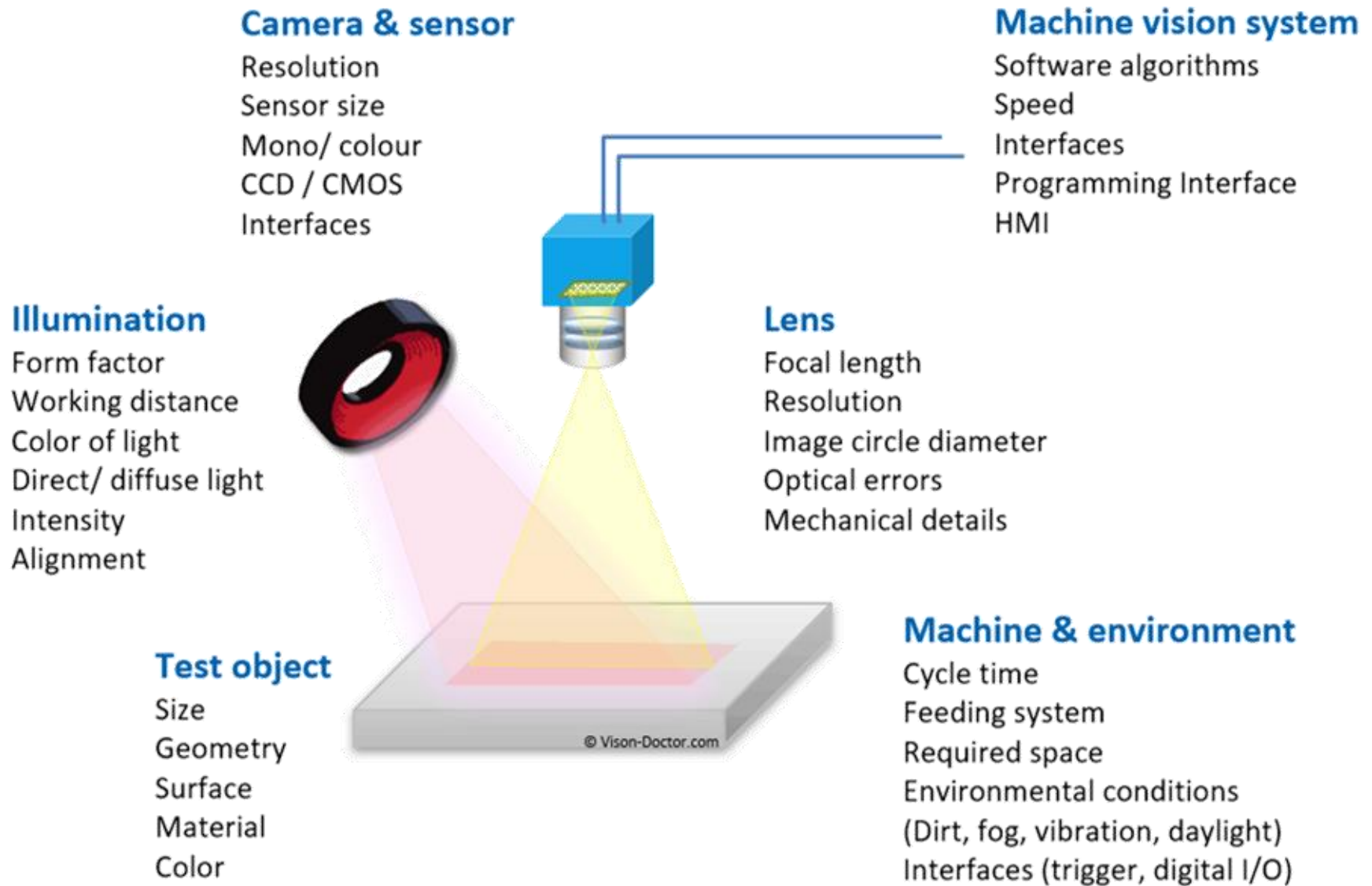
Closed Loop Control System

- What are we discussing in this Segment?



- Feedback elements / sensors
 - Vision sensors
 - Event-based Vision Sensors (EVS)

Conventional Machine Vision (CIS-based) Systems



Sensor Functions Inside a Camera

- Typical CMOS Camera (CIS) Layout

Sensor Functions:

1. Photoelectric Conversion

Converts photons into electrons

2. Charge Accumulation

Collects generated charge as signal charge

3. Transfer Signal

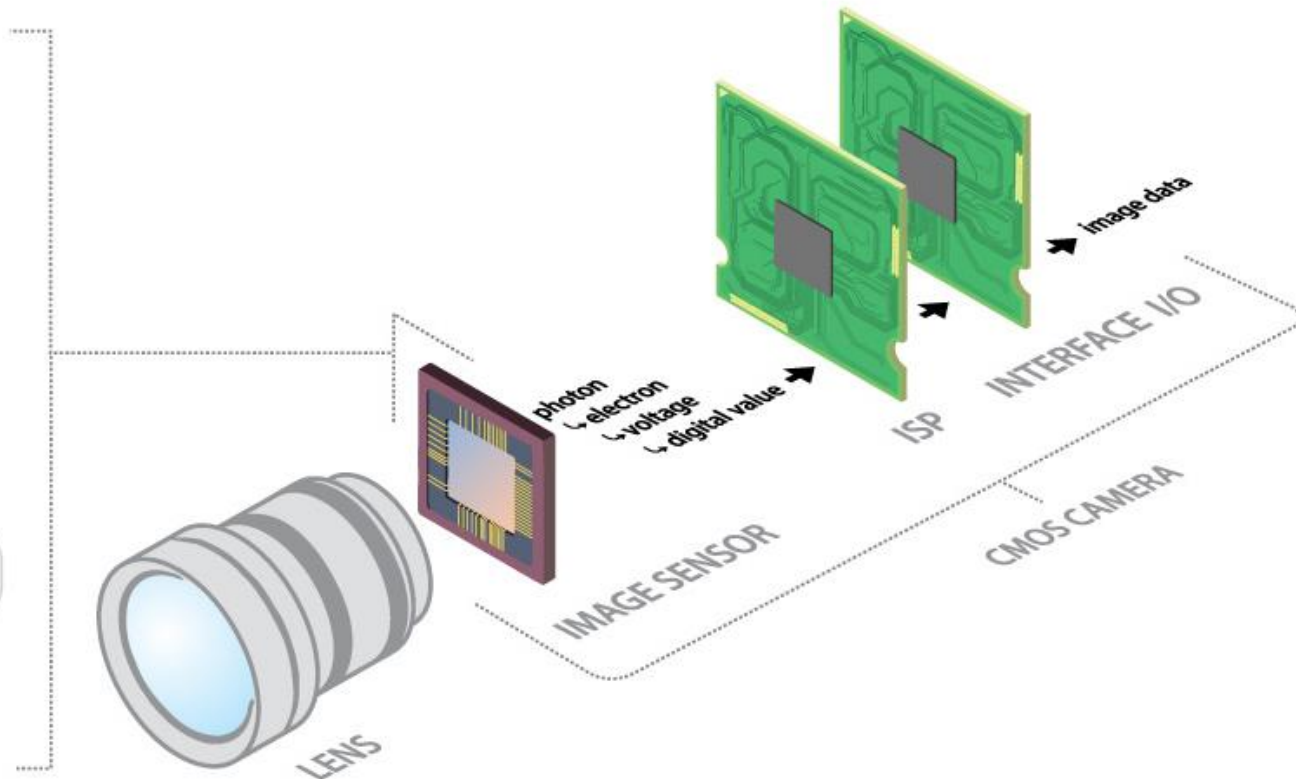
Moves signal charge to detecting node

4. Signal Detection

Converts signal charge into electrical signal (voltage)

5. Analog to Digital Conversion

Converts voltage into digital value



Open Challenges in Computer Vision

The past 60 years of research have been devoted to frame-based cameras ...but they are not good enough!

Latency & Motion blur



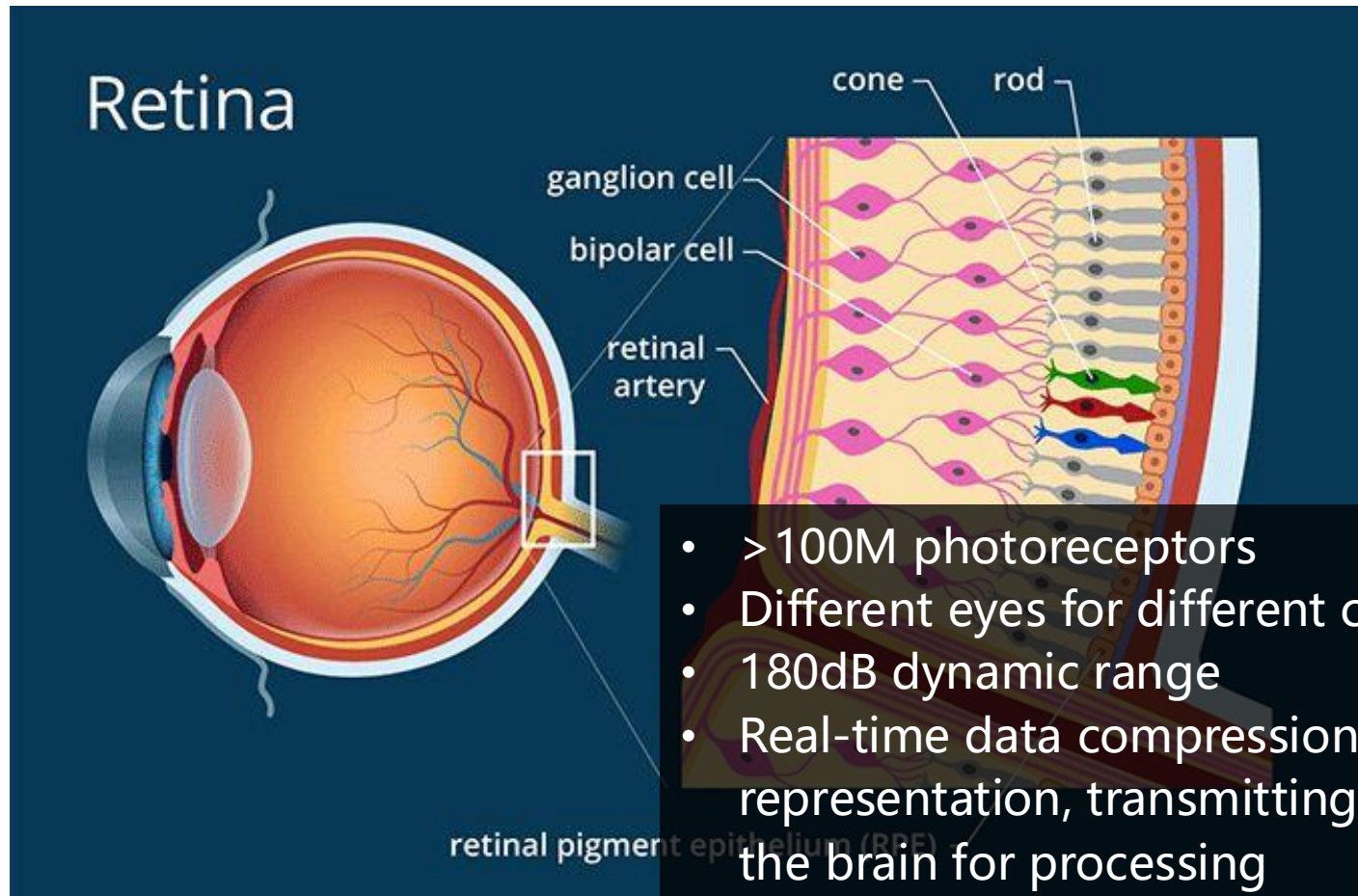
Dynamic Range



Event cameras do not suffer from these problems!

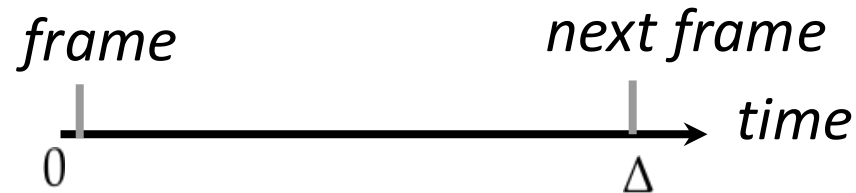
Inspiration

- Inspired by biological sensor – our eyes

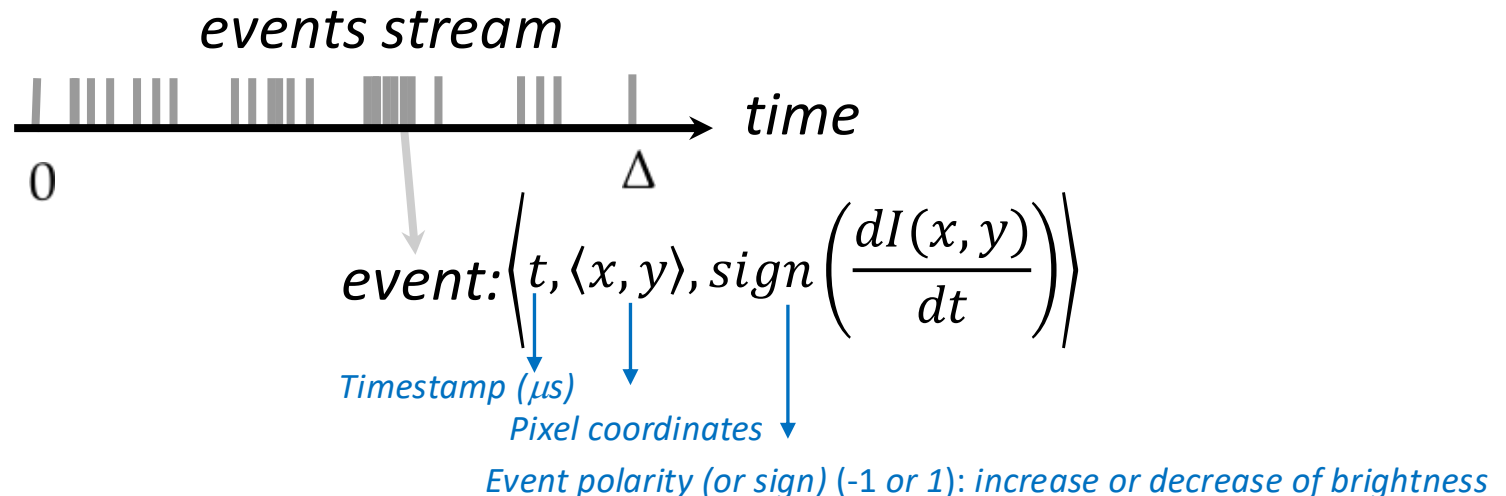


Traditional Camera vs. Event Camera

- A traditional camera outputs frames at fixed time intervals:



- By contrast, an event camera outputs asynchronous events at microsecond resolution. An **event** is generated each time a single pixel detects an intensity change that is larger than the threshold.



What is an Event-based Vision Sensor (EVS)?

- Novel sensor that measures only **pixel-wise brightness change**
- **First commercialized in 2008** by T. Delbruck (UZHÐ) under the name of Dynamic Vision Sensor (DVS)
- **Low-latency** ($\sim 1 \mu\text{s}$)
- **No motion blur**
- **High dynamic range** (140 dB instead of 60 dB)
- **Ultra-low power** (mean: 1mW vs 1W)

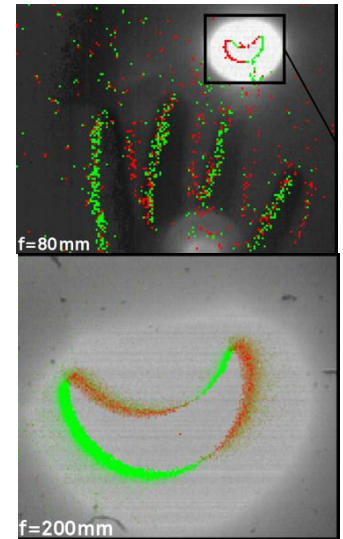
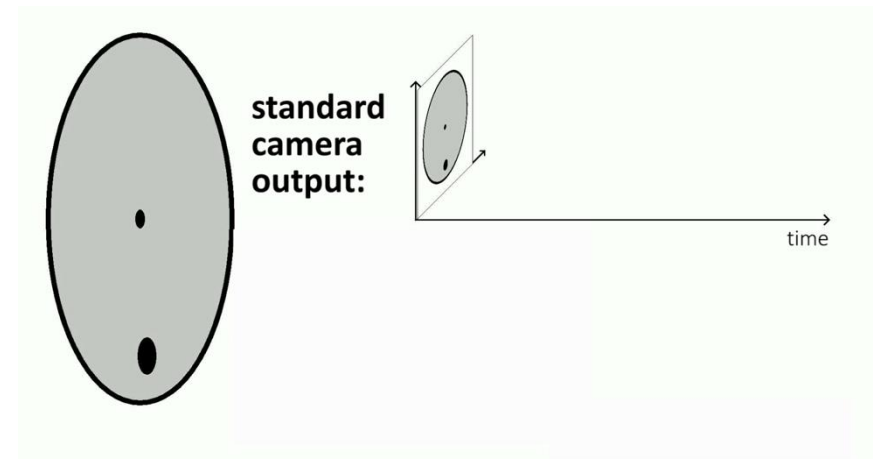


Image of the solar eclipse captured by a DVS

Traditional vision algorithms cannot be used because:

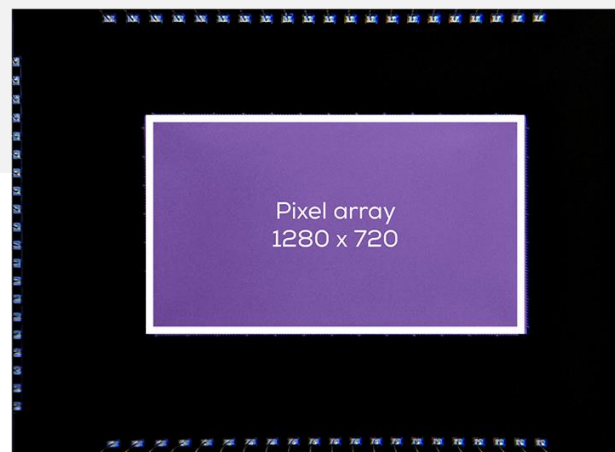
- **Asynchronous** pixels
- **No intensity information** (only binary intensity changes)



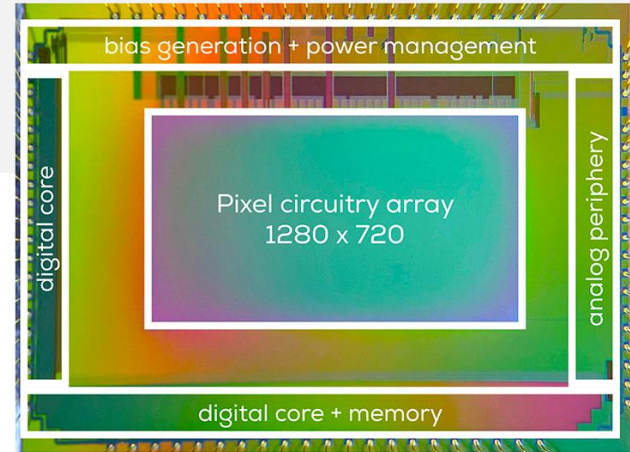
[Lichtsteiner, Posch, Delbruck, [A 128x128 120 dB 15 \$\mu\$ s Latency Asynchronous Temporal Contrast Vision Sensor](http://rpg.ifi.uzh.ch/docs/scaramuzza/Tutorial_on_Event_Cameras_Scaramuzza.pdf), 2008
http://rpg.ifi.uzh.ch/docs/scaramuzza/Tutorial_on_Event_Cameras_Scaramuzza.pdf]

EVS Mechanism

- Chip photographs



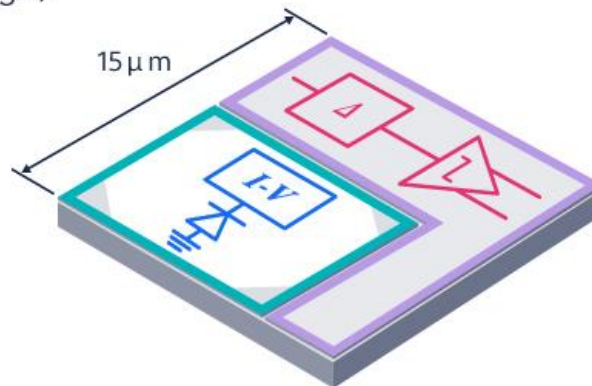
Micrograph of bottom part



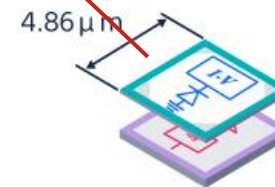
Micrograph of top part

- Stacked layers: 77% fill factor

〈Pixel image〉



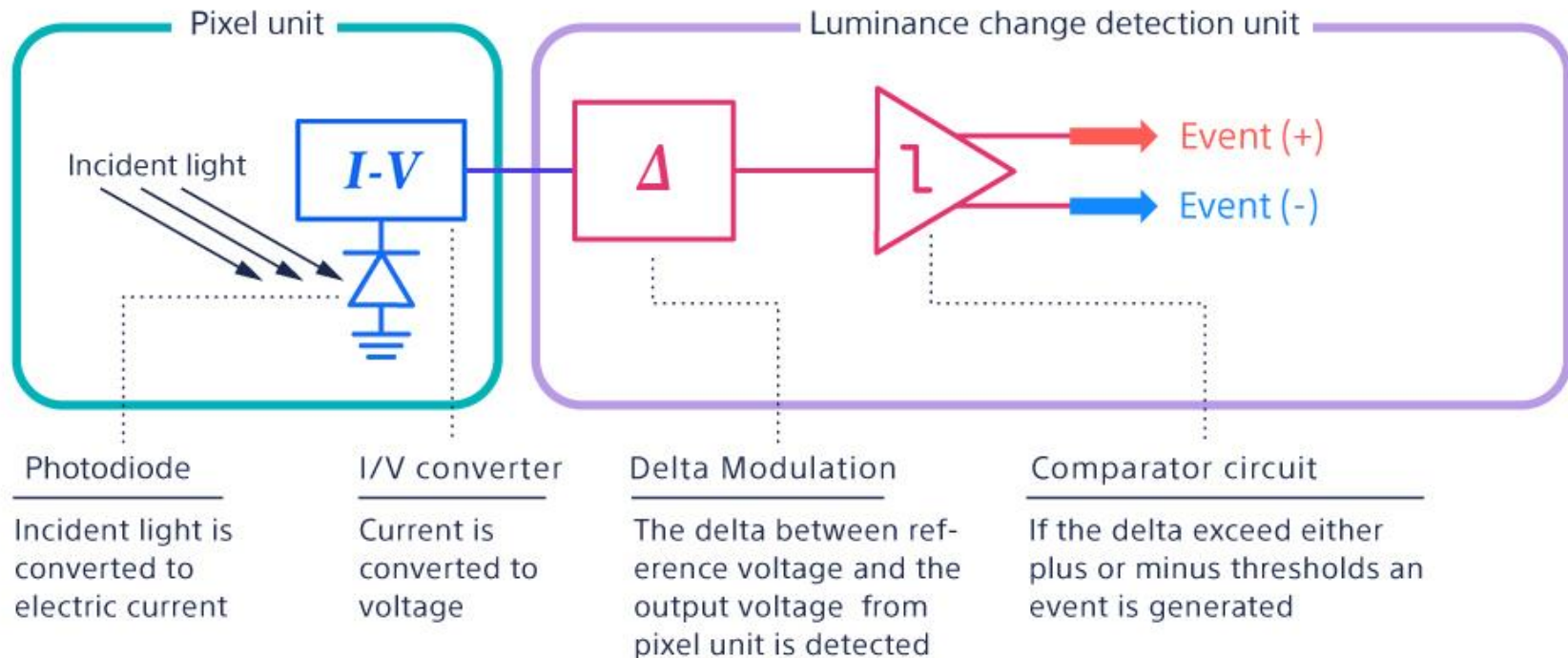
Conventional sensor (Front-side illuminated type)



This work

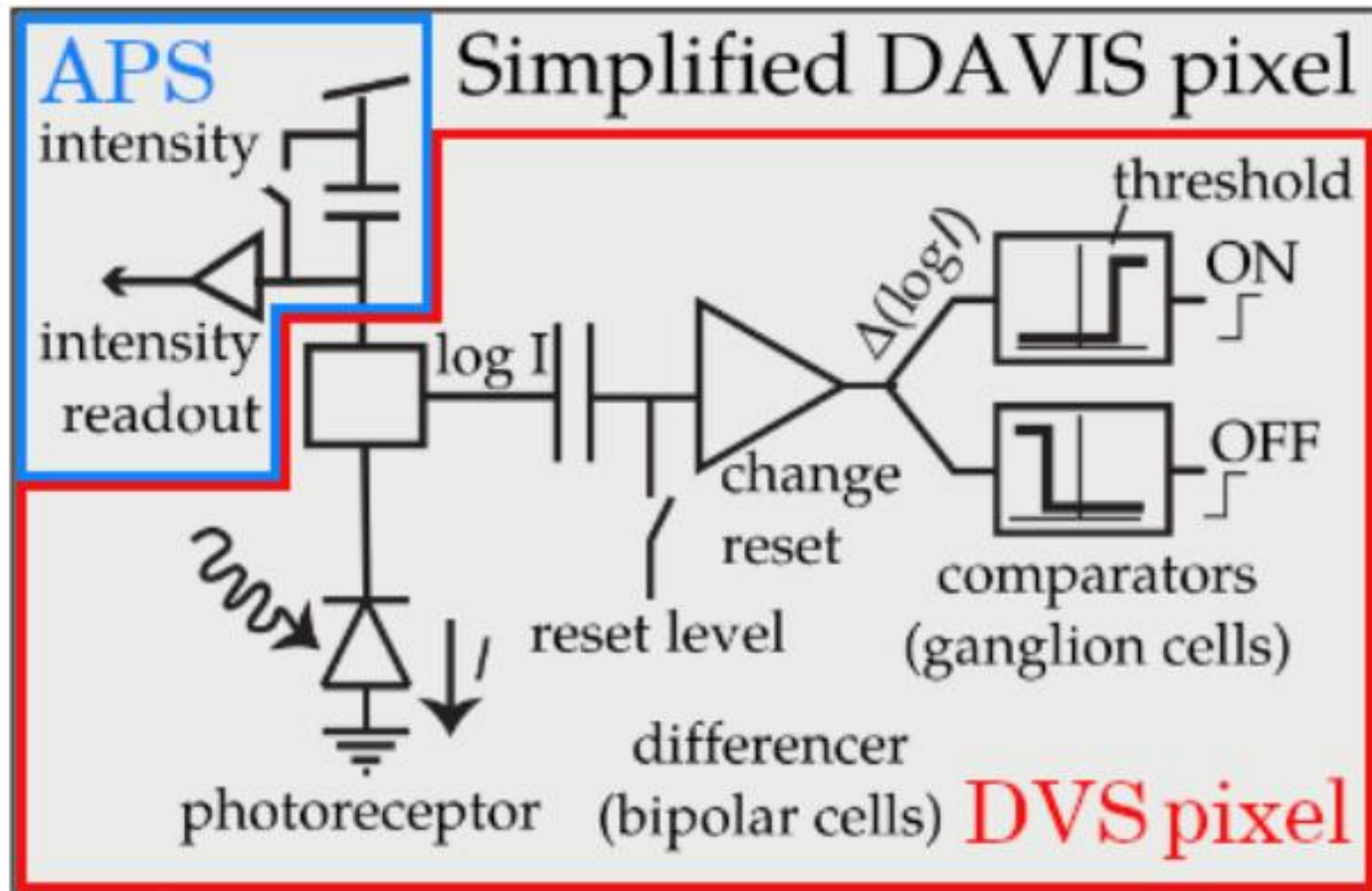
EVS Mechanism

- Pixel diagram of EVS



EVS Mechanism

- Pixel diagram of EVS **with absolute intensity measurement**



EVS Mechanism

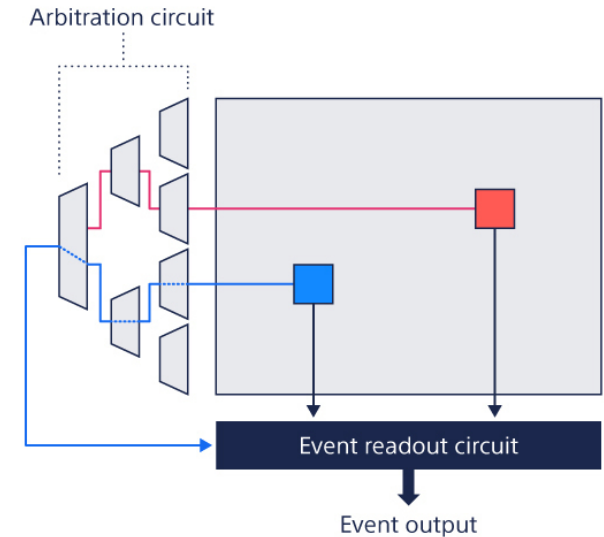
- Event read-out: aims to achieve highspeed and lower latency by either
 - Arbitration circuit (in right fig.)

Pros and cons?

- Row and column scanner and read-out

Pros and cons?

- Built-in hardware event filter and event signal processing (ESP)
 - Denoiser
 - Anti-flickering



[Event-based Vision Sensor \(EVS\) Technology | Image Sensor for Industrial Use | Technology | Sony Semiconductor Solutions Group \(sony-semicon.com\)](#)

[Prophesee and Sony Develop a Stacked Event-Based Vision Sensor](#)
[CVPRW2023 EVS presentation 230619.pptx - Google Slides](#)

EVS Mechanism

- Example: a CIS-EVS hybrid 3-layer stacking sensor

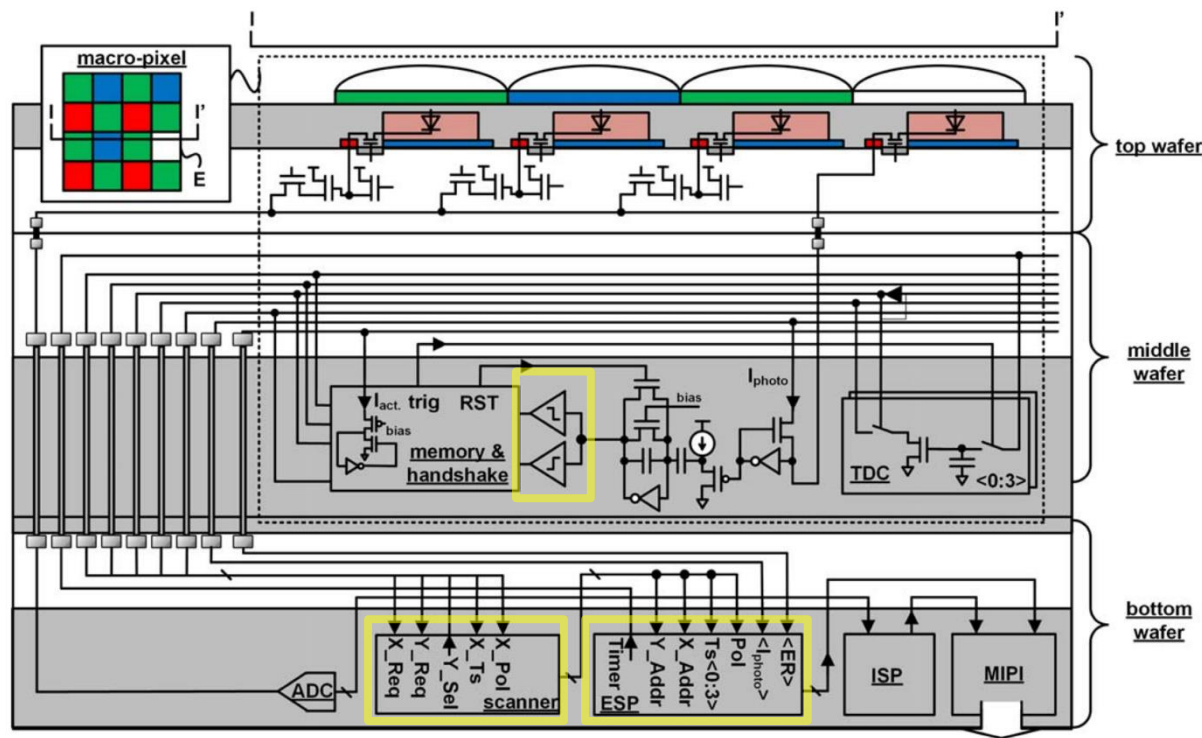
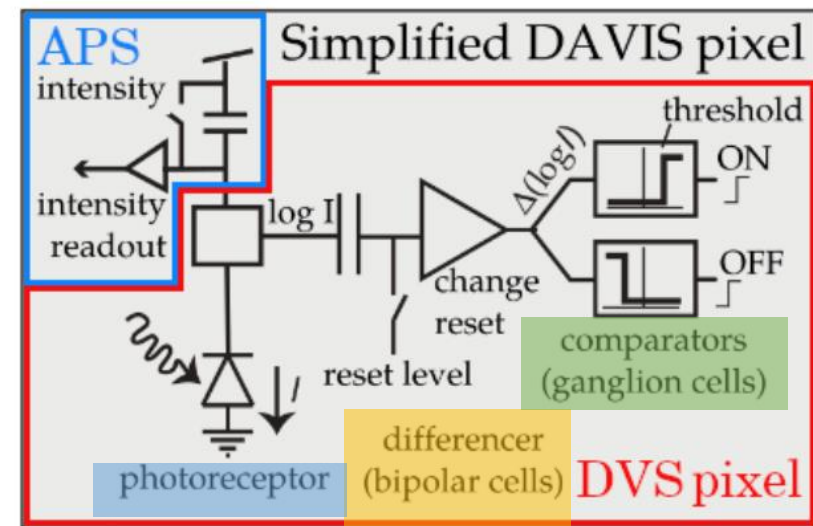


Figure 5.1.1: Schematic of dual wafer 4×4 macro-pixel and peripheral readout circuitry on third wafer.

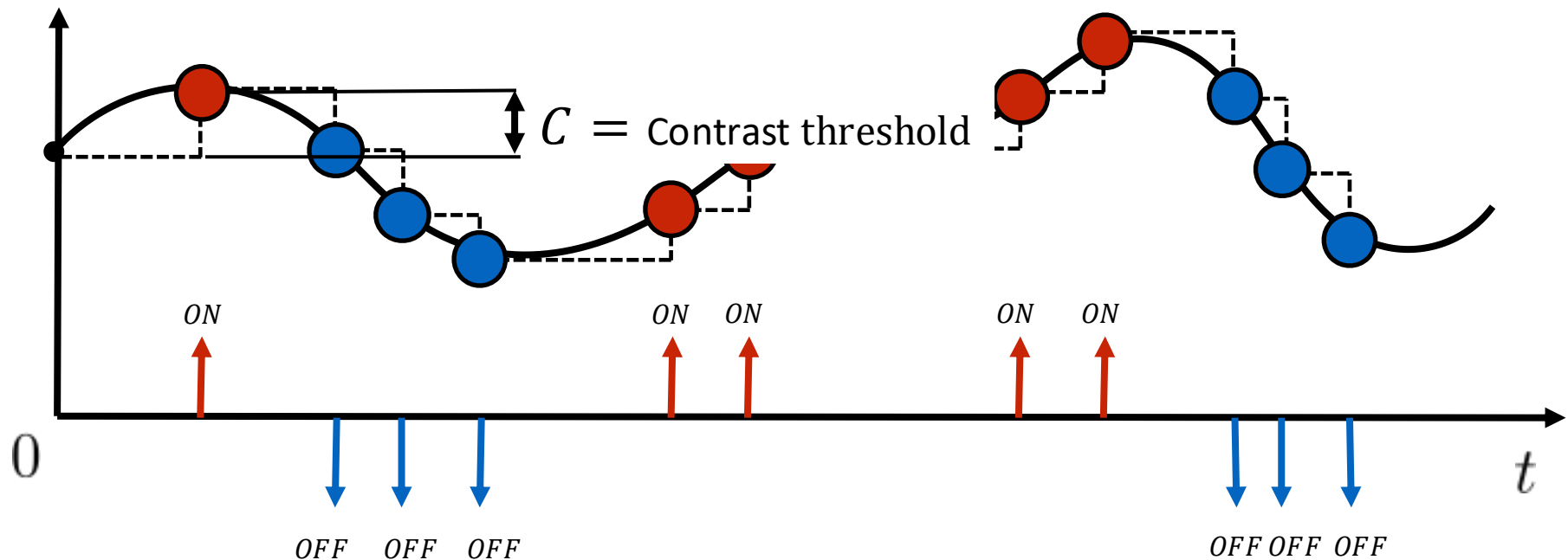
Generative Event Model

Why events are generated?

- Consider the intensity at a single pixel...
$$\pm C = \log I(x, t) - \log I(x, t - \Delta t)$$



$\log I(x, t)$

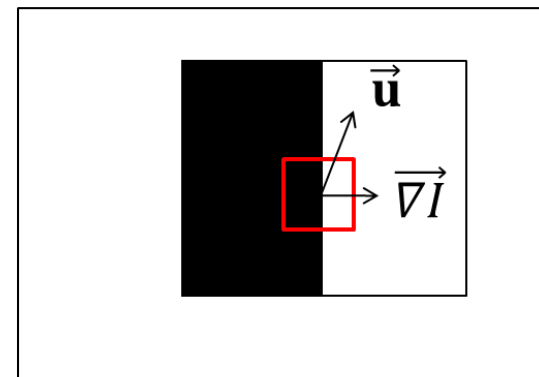
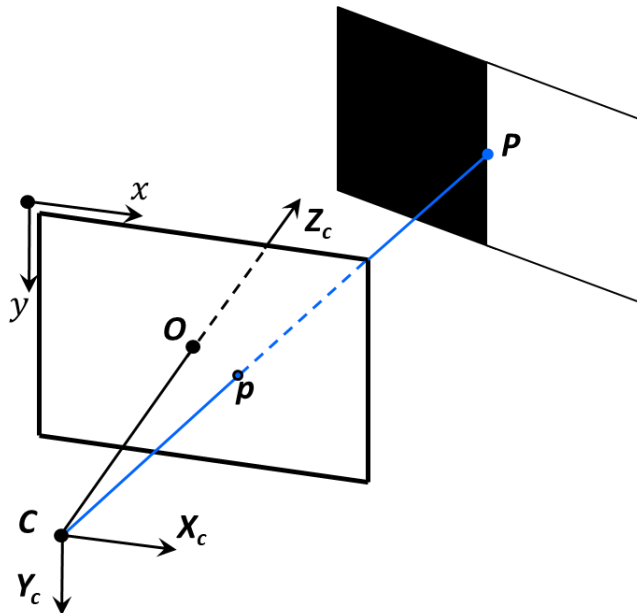


Events are triggered **asynchronously!**

Generative Event Model: 1st Order Approximation

- Let us define $L(x,y,t) = \text{Log}(I(x,y,t))$
- Consider a given pixel $p(x,y)$ with gradient $\nabla L(x,y)$ undergoing the motion $\mathbf{u} = (u,v)$ in pixels, induced by a moving 3D point $\mathbf{P}$.
- Then, it can be shown that:

$$-\nabla L \cdot \mathbf{u} = C$$



Generative Event Model: 1st Order Approximation – Proof

- Recall the **Brightness constancy assumption**, which says: the intensity value of p , **before and after the motion**, must **remain unchanged**.

$$L(x, y, t) = L(x + u, y + v, t + \Delta t)$$

- By replacing the right-hand term by its 1st order approximation at $t + \Delta t$:

$$L(x, y, t) = L(x, y, t + \Delta t) + \frac{\partial L}{\partial x} u + \frac{\partial L}{\partial y} v$$

$$\Rightarrow L(x, y, t + \Delta t) - L(x, y, t) = -\frac{\partial L}{\partial x} u - \frac{\partial L}{\partial y} v$$

$$\Rightarrow \Delta L = \boxed{C = -\nabla L \cdot \mathbf{u}}$$

- This equation describes the **linearized** event generation equation for an event generated by a gradient ∇L that moved by a motion vector $\mathbf{u}$ (optical flow) during a time interval Δt .

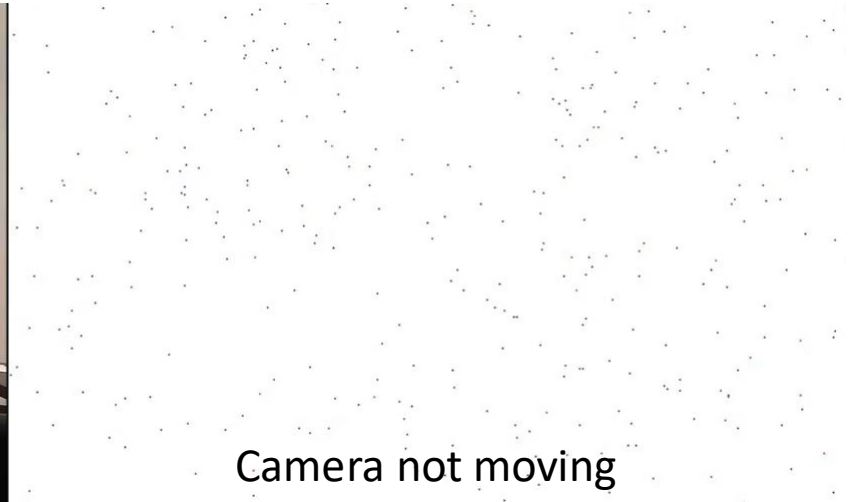
Event Camera Output With/Without Motion

- Without motion, only output is the background noise

Standard Camera



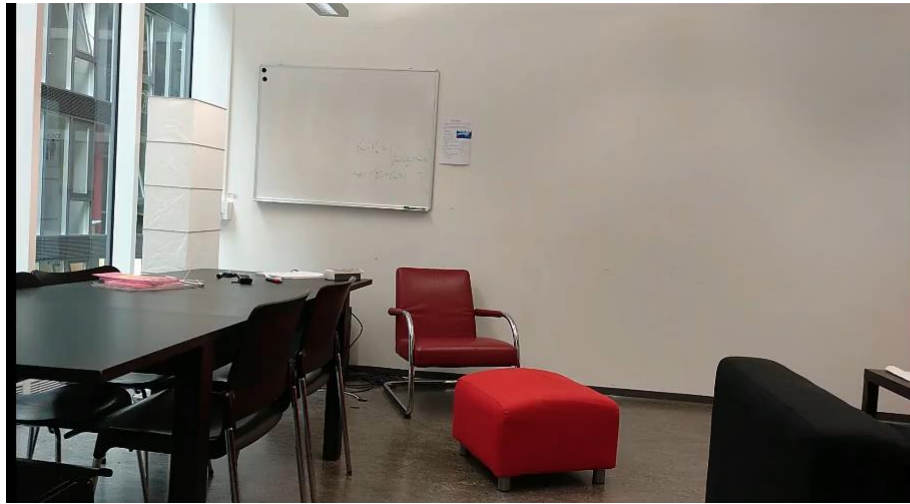
Event Camera (ON, OFF events)



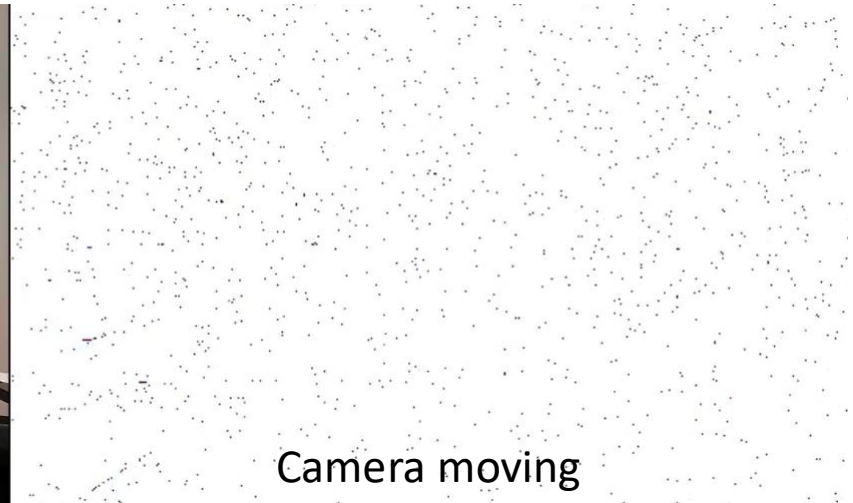
Camera not moving

Event Camera Output With/Without Motion

Standard Camera

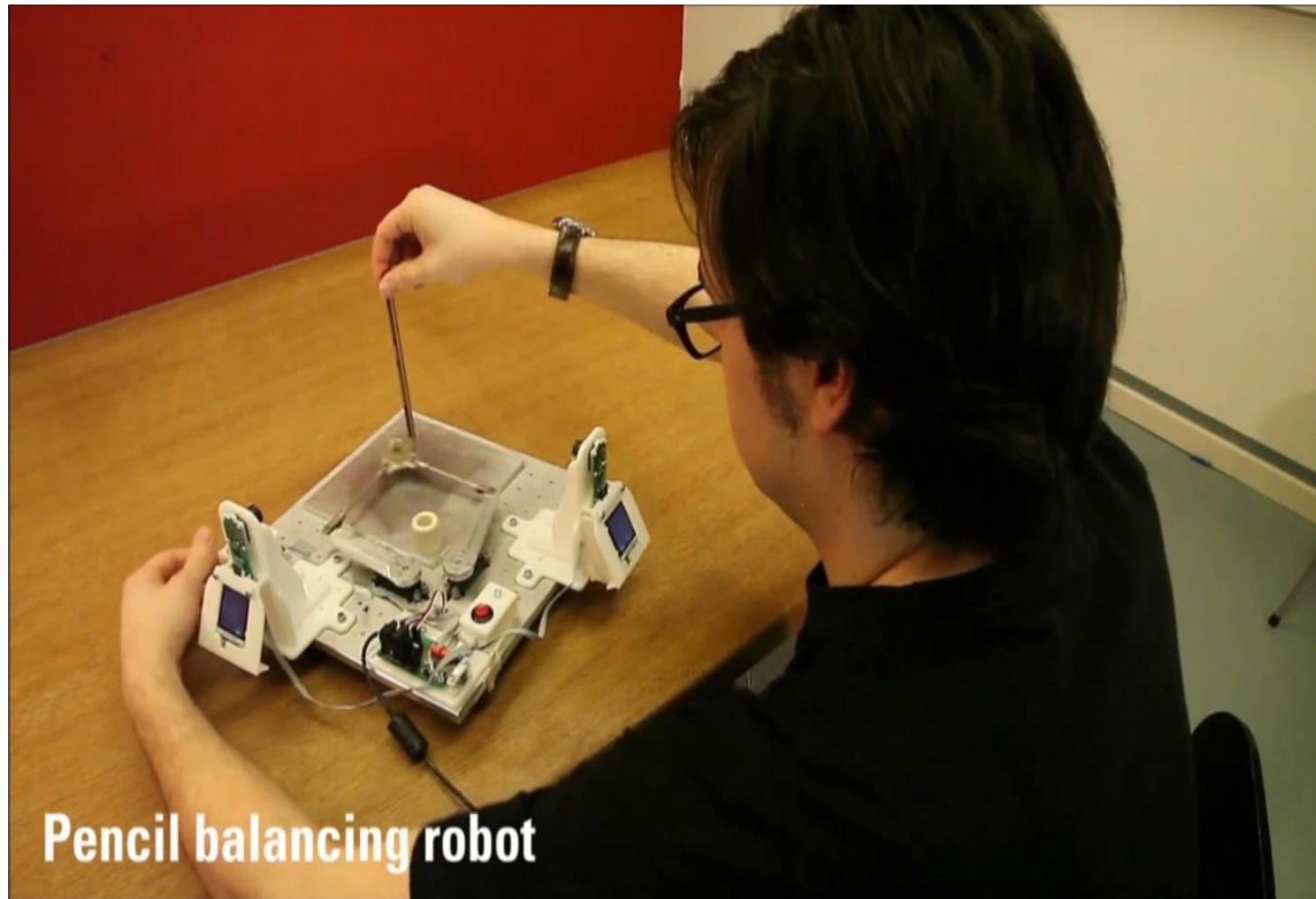


Event Camera (ON, OFF events)



Examples: Inverted Pendulum

- Low latency, high-speed measurement on pencil angle using event cameras



Pencil balancing robot

Conradt, Cook, Berner, Lichtsteiner, Douglas, Delbruck, **A pencil balancing robot using a pair of AER dynamic vision sensors**, IEEE International Symposium on Circuits and Systems. 2009

Examples: Low-light Sensitivity (Night Drive)

- Excellent night vision ability



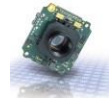
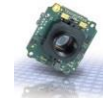
GoPro Hero 6



Event Camera by Prophesee
White = Positive events
Black = Negative events

High-speed vs. Event Cameras

- Can I just use a high-speed camera?

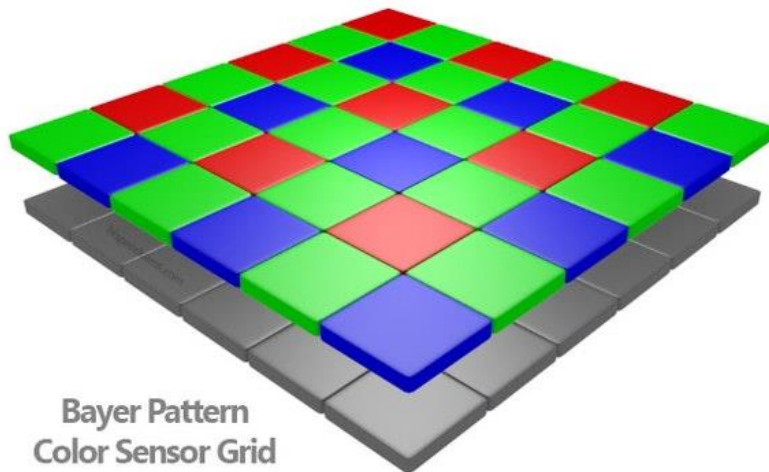


	High speed camera	Standard camera	Event Camera
Max fps or measurement rate	Up to 1MHz	100-1,000 fps	1MHz
Resolution at max fps	64x16 pixels	>1Mpxl	>1Mpxl
Bits per pixels/ Bits per event	12 bits/pixel	8-10bits/pixel	~40 bits/event $\{t, (x, y), p\}$
Weight	6.2 kg	30 g	30 g
Active cooling	yes	No cooling	No cooling
Data rate	1.5 GB/s	32MB/s	~1MB/s on average (depends on dynamics)
Mean power consumption	150 W + external light	1 W	1 mW
Dynamic range	n.a.	60 dB	140 dB

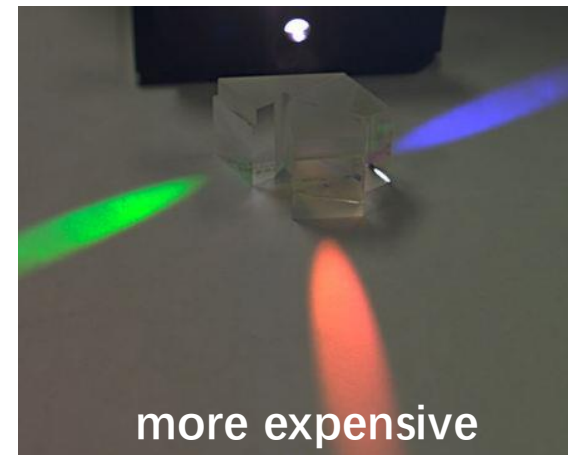
Color Event Cameras

- Two ways to build a color event camera
 - color filter
 - color-splitter prisms
- Some publications of practical color event cameras available
- Simulated data available for research

color filters



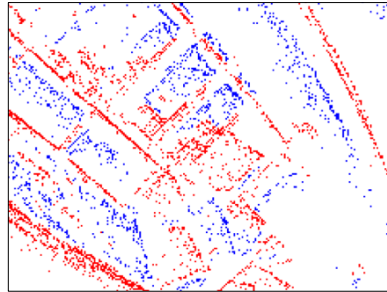
color-splitter prisms



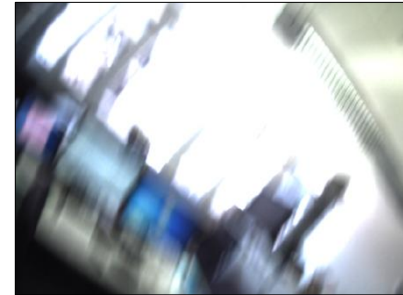
<https://www.youtube.com/watch?v=CysnFwFetpM>

Event Camera + Conventional Camera + ...

- **Design idea:** utilize the complementary advantages of event and standard cameras, and other sensors.



Event Camera



Standard Camera

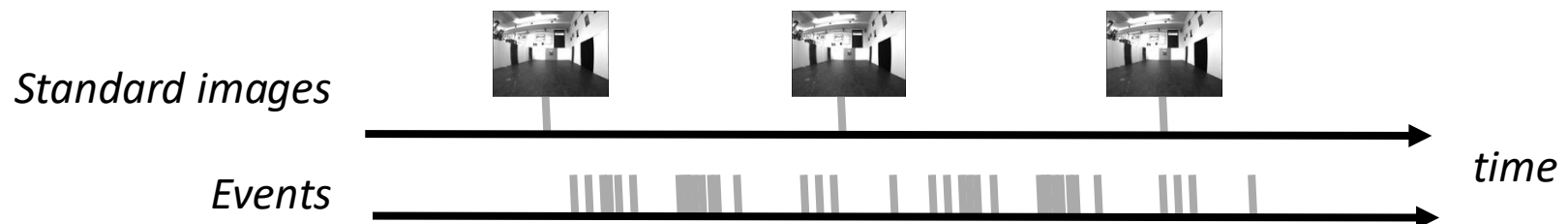
Update rate	High (asynchronous): 1 MHz	Low (synchronous)
Dynamic Range	High (140 dB)	Low (60 dB)
Motion Blur	No	Yes
Static motion	No (event camera is a high pass filter)	Yes
Absolute intensity	No (reconstructable up to a constant)	Yes
Maturity	~ 10 years of research	> 60 years of research!

Example: DAVIS Sensor (Events + Images + IMU)

- Combines an event and a standard camera in the **same pixel array** (→ the same pixel can both trigger events and integrate light intensity).
- It also has an IMU.



Temporal aggregation of events
overlaid on a DAVIS frame



Event Camera Manufactures and Products



Supplier		iniVation			Prophesee				Samsung			CelePixel		Insightness
Camera model		DVS128	DAVIS240	DAVIS346	ATIS	Gen3 CD	Gen3 ATIS	Gen 4 CD	DVS-Gen2	DVS-Gen3	DVS-Gen4	CeleX-IV	CeleX-V	Rino 3
Sensor specifications	Year, Reference	2008 [2]	2014 [4]	2017	2011 [3]	2017 [67]	2017 [67]	2020 [68]	2017 [5]	2018 [69]	2020 [39]	2017 [70]	2019 [71]	2018 [72]
	Resolution (pixels)	128 × 128	240 × 180	346 × 260	304 × 240	640 × 480	480 × 360	1280 × 720	640 × 480	640 × 480	1280 × 960	768 × 640	1280 × 800	320 × 262
	Latency (μs)	12μs @ 1klux	12μs @ 1klux	20	3	40 - 200	40 - 200	20 - 150	65 - 410	50	150	10	8	125μs @ 10lux
	Dynamic range (dB)	120	120	120	143	> 120	> 120	> 124	90	90	100	90	120	> 100
	Min. contrast sensitivity (%)	17	11	14.3 - 22.5	13	12	12	11	9	15	20	30	10	15
	Power consumption (mW)	23	5 - 14	10 - 170	50 - 175	36 - 95	25 - 87	32 - 84	27 - 50	40	130	-	400	20-70
	Chip size (mm ²)	6.3 × 6	5 × 5	8 × 6	9.9 × 8.2	9.6 × 7.2	9.6 × 7.2	6.22 × 3.5	8 × 5.8	8 × 5.8	8.4 × 7.6	15.5 × 15.8	14.3 × 11.6	5.3 × 5.3
	Pixel size (μm ²)	40 × 40	18.5 × 18.5	18.5 × 18.5	30 × 30	15 × 15	20 × 20	4.86 × 4.86	9 × 9	9 × 9	4.95 × 4.95	18 × 18	9.8 × 9.8	13 × 13
	Fill factor (%)	8.1	22	22	20	25	20	> 77	11	12	22	8.5	8	22
	Supply voltage (V)	3.3	1.8 & 3.3	1.8 & 3.3	1.8 & 3.3	1.8	1.8	1.1 & 2.5	1.2 & 2.8	1.2 & 2.8		1.8 & 3.3	1.2 & 2.5	1.8 & 3.3
	Stationary noise (ev/pix/s) at 25C	0.05	0.1	0.1	-	0.1	0.1	0.1	0.03	0.03		0.15	0.2	0.1
	CMOS technology (nm)	350	180	180	180	180	180	90	90	90	65/28	180	65	180
		2P4M	1P6M MIM	1P6M MIM	1P6M	1P6M CIS	1P6M CIS	BI CIS	1P5M BSI			1P6M CIS	CIS	1P6M CIS
Camera	Grayscale output	no	yes	yes	yes	no	yes	no	no	no	no	yes	yes	yes
	Grayscale dynamic range (dB)	NA	55	56.7	130	NA	> 100	NA	NA	NA	NA	90	120	50
	Max. frame rate (fps)	NA	35	40	NA	NA	NA	NA	NA	NA	NA	50	100	30
Camera	Max. Bandwidth (Meps)	1	12	12	-	66	66	1066	300	600	1200	200	140	20
	Interface	USB 2	USB 2	USB 3		USB 3	USB 3	USB 3	USB 2	USB 3	USB 3			USB 2
	IMU output	no	1 kHz	1 kHz	no	1 kHz	1 kHz	no	no	1 kHz	no	no	no	1 kHz

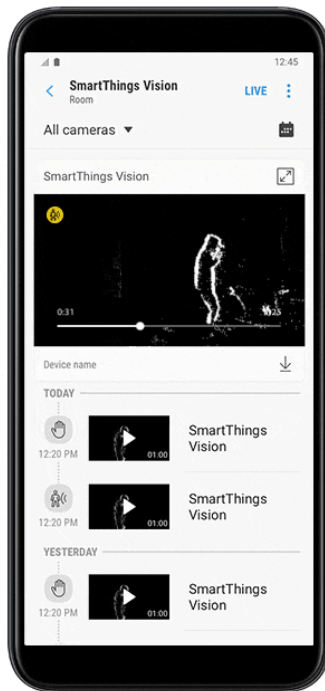
Event Camera Manufactures and Products

- SmartThings Vision by Samsung

“Security that respects your privacy”

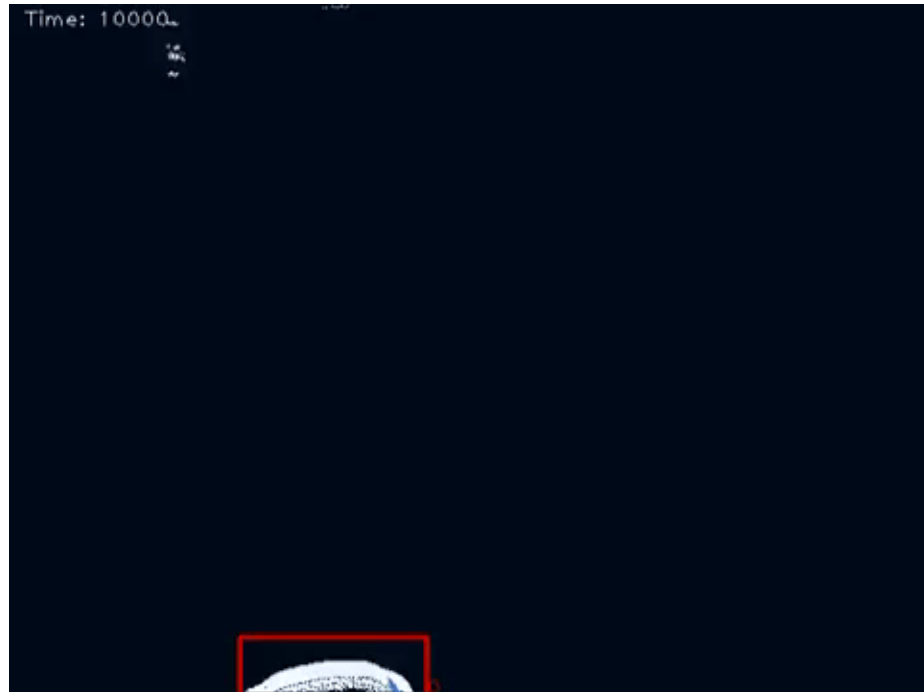
Unexpected person
or intruder warning

while helps protects the
privacy



Event Camera Manufactures and Products

- Industrial solutions by Phophesee
 - ULTRA SLOW MOTION - Up to 200,000 fps (time resolution equivalent)
 - OBJECT TRACKING - Continuous tracking in time, no blind spots between frames



Event Camera Manufactures and Products

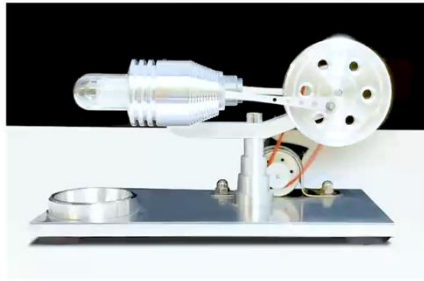
- Industrial solutions by Phophesee
 - OPTICAL FLOW – intrinsic from the sensor, much less power consumption
 - HIGH-SPEED COUNTING – Throughput $> 1,000$ Obj/s
Accuracy $> 99.5\%$ @ $1,000$ Obj/s.



PROPHESÉE
META VISION FOR MACHINES

Event Camera Manufactures and Products

- Industrial, IoT, and robotics solutions by Phophesee
 - VIBRATION MONITORING - From 1Hz to kHz range
 - VISUAL ODOMETRY – under fast motion



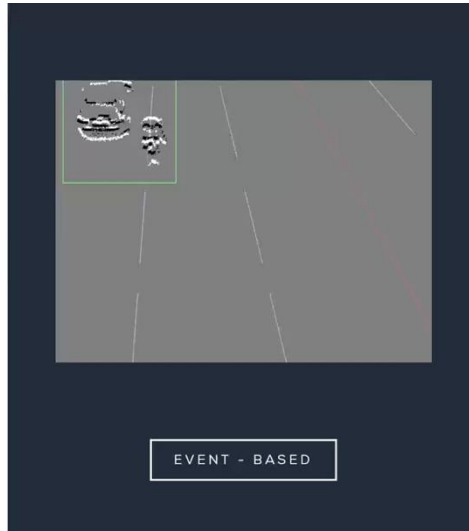
PROPHESÉE
META VISION FOR MACHINES

Event Camera Manufactures and Products

- Industrial, IoT, and robotics solutions by Phophesee
 - TRAFFIC DATA ACQUISITION – vehicle speed measurement
 - GESTURE DETECTION – for human-machine interaction



FRAME - BASED



EVENT - BASED



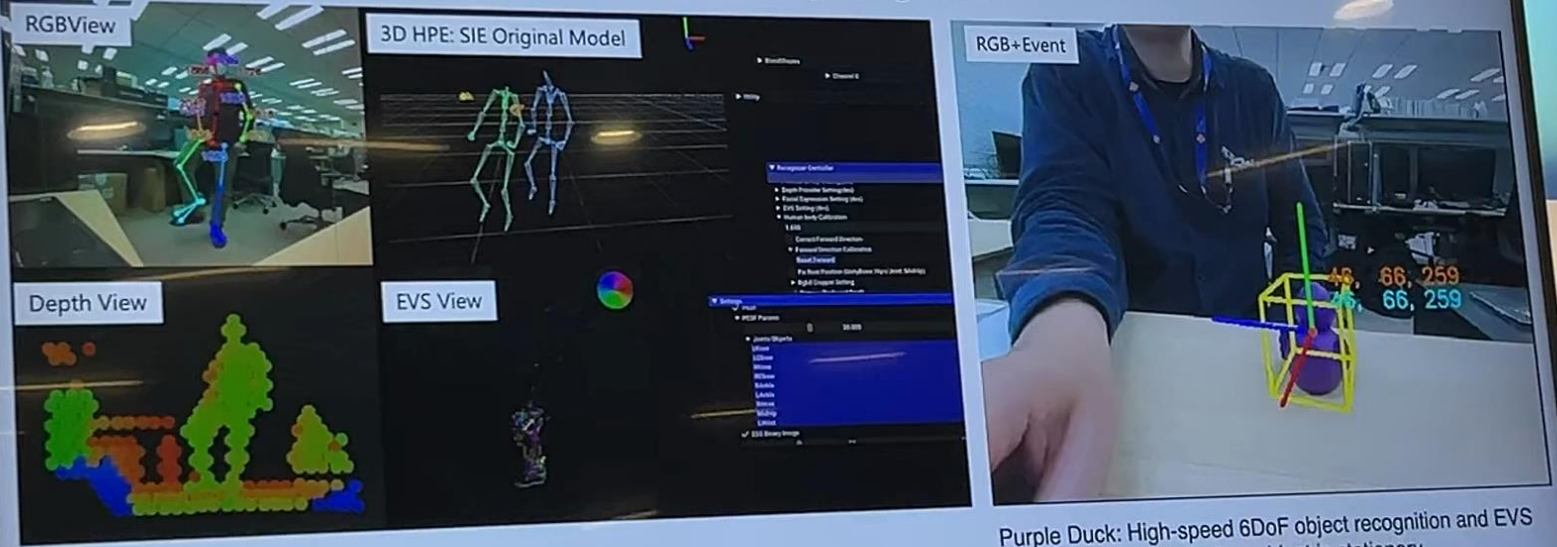
Event Camera Manufactures and Products

- Gaming solutions by SONY

RGB-DE Human + 6DoF Object Pose Estimation

FTG Future Technology Group

- Real-time 3D Human Pose Estimation demo using RGB-DE with one camera unit
- Integration of Object 6DoF Recognition Technology using RGB-E



RGBView

3D HPE: SIE Original Model

Depth View

EVS View

RGB+Event

Purple Duck: High-speed 6DoF object recognition and EVS
Perfect jitter removal when object is stationary
Orange : RGB recognized coordinates
Light blue : RGB+EVS filtered coordinates

RGB 2D human pose results combined with Depth
Convert to 3D human body posture, EVS for fast movements
Optical Flow Tracking

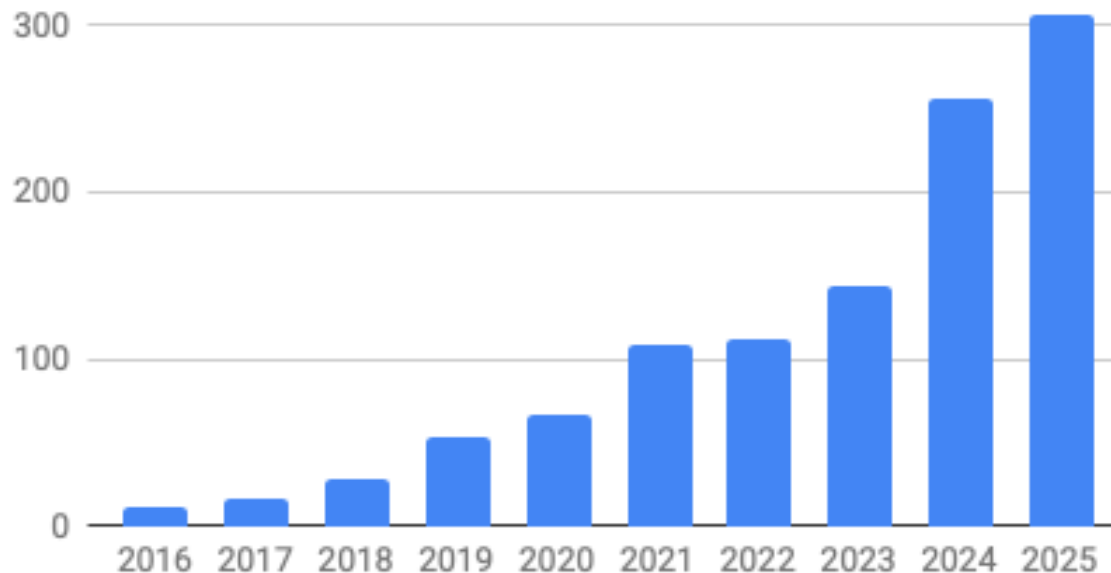
©2024 Sony Interactive Entertainment

7

Fast-Growing Community

- Statistics: Papers in computer vision and robotics venues

Papers in Computer Vision and Robotics venues



- Event-based Vision Resources:

[GitHub - uzh-rpg/event-based_vision_resources](https://github.com/uzh-rpg/event-based_vision_resources)

Summary – Event-based Sensors and Cameras

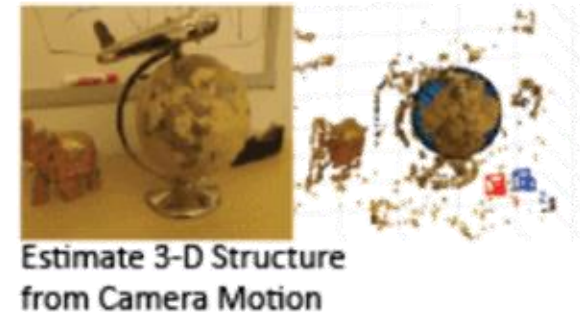
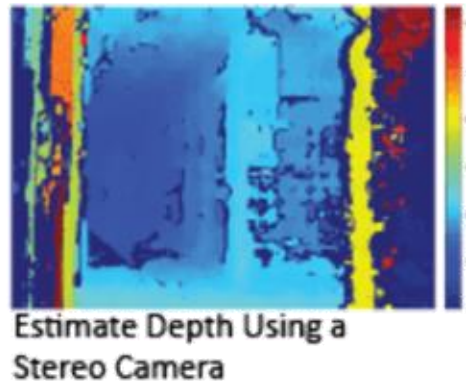
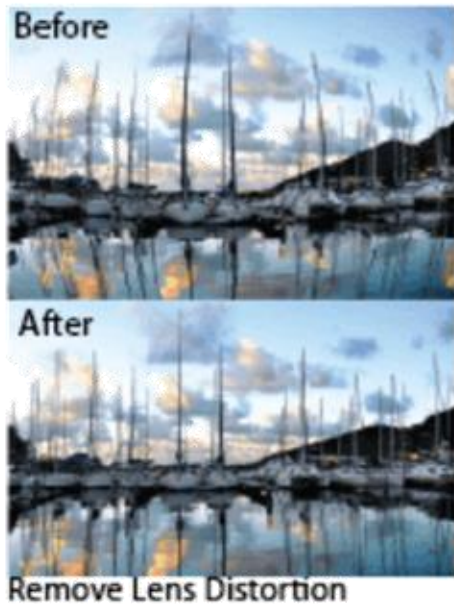
- Why we need event cameras: the motivation
- EVS principles and mechanisms
 - Pixel design
 - The generative event model
- Event camera manufactures, products, and applications
- Fast-growing EVS community

What else we need to build an event-based vision system?

Break (15 mins)

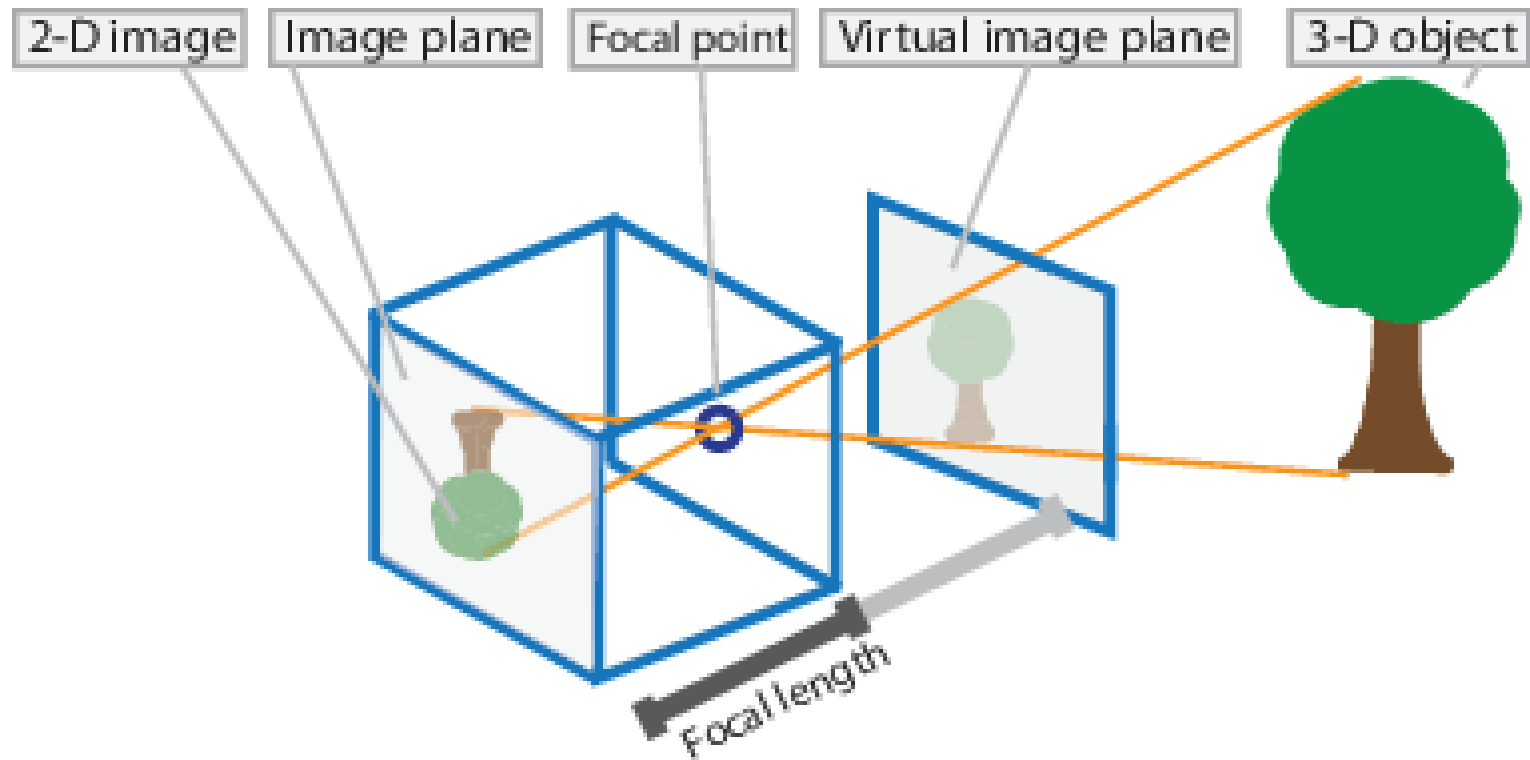
Event Camera Calibration

- Why need camera calibration?
 - To estimate the **parameters** of a lens and image sensor.
- With a calibrated camera, you can:



Camera Models

- Pinhole camera model
 - A simple camera without a lens and with a single small aperture



Camera Models

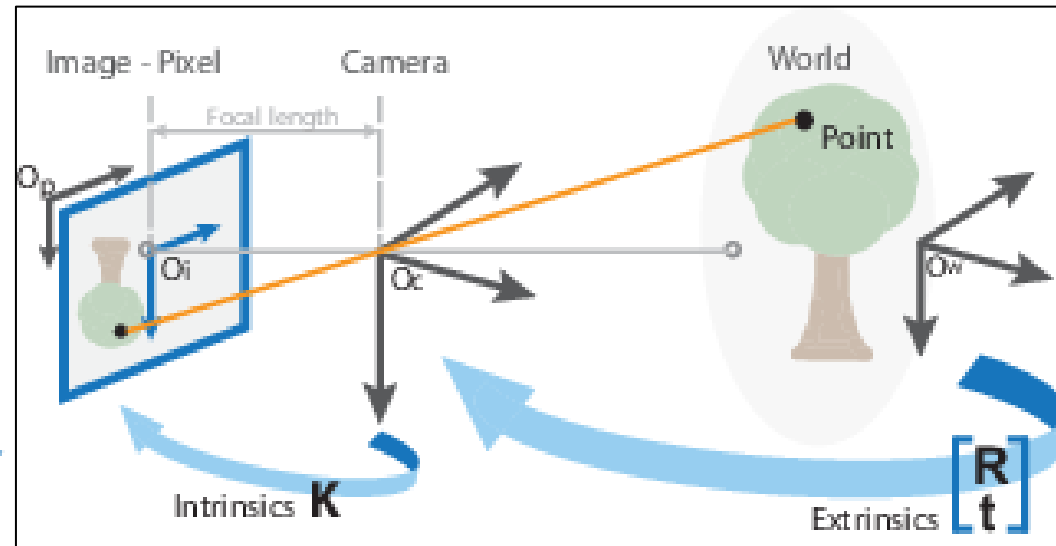
- Pinhole camera model
 - Parameters are included in a 4-by-3 **camera matrix**: **3-D** $\rightarrow$ **2-D**

$$w \begin{bmatrix} x & y & 1 \end{bmatrix} = \begin{bmatrix} X & Y & Z & 1 \end{bmatrix} P$$

Scale factor *Image points* *World points*

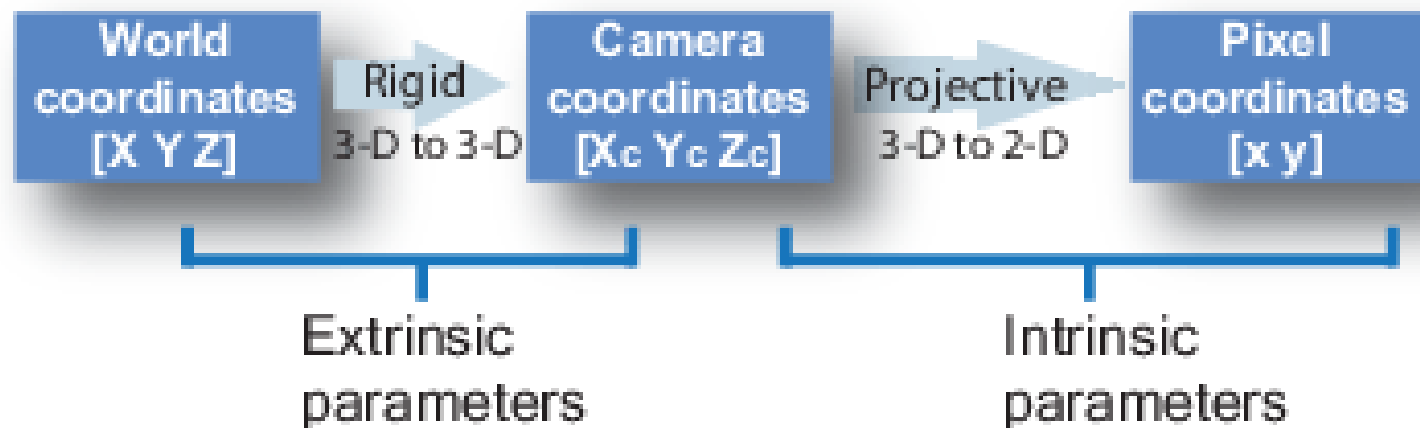
$$P = \begin{bmatrix} R \\ t \end{bmatrix} K$$

Camera matrix *Extrinsics* *Intrinsic matrix*
Rotation and translation



Camera Models

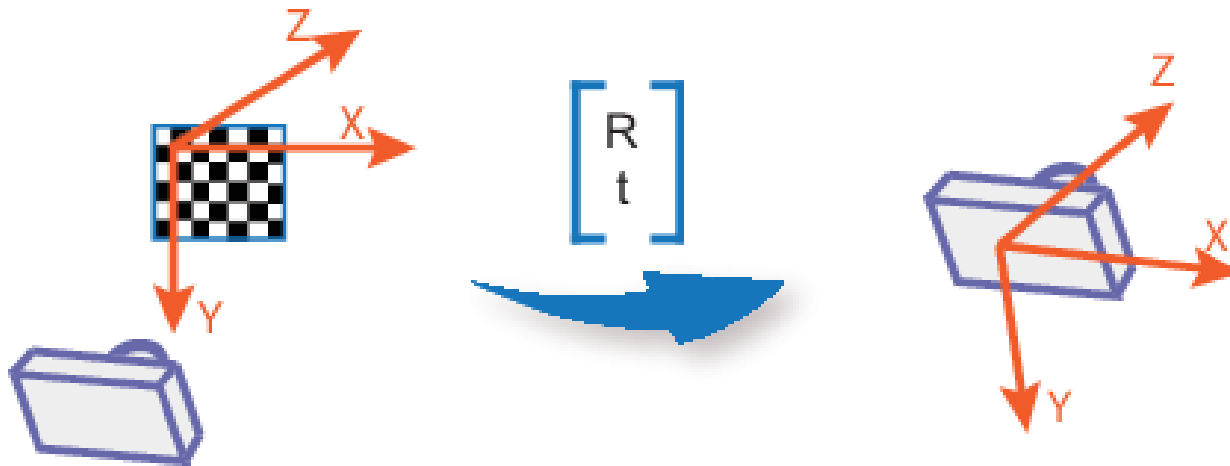
- Pinhole camera model
 - Extrinsic parameters: **world points** → **camera coordinates**
 - Intrinsic parameters: **camera coordinates** → **image plane**



With extrinsics and intrinsic parameters, we could obtain the **camera matrix**!

Camera Models

- Extrinsic parameters
 - consist of a **rotation**, $\mathbf{R}$, and a **translation**, $\mathbf{t}$.
- $\mathbf{R}$: rotation matrix
 $\mathbf{t}$: translation vector



Camera Models

- Intrinsic parameters
 - include: the **focal length**, the **optical center**, also known as the **principal point**, and the **skew coefficient**

$$\begin{bmatrix} f_x & 0 & 0 \\ s & f_y & 0 \\ c_x & c_y & 1 \end{bmatrix}$$

What is the unit of c_x and c_y ?

$[c_x \ c_y]$ – Optical center (the principal point), in pixels.

(f_x, f_y) – Focal length in pixels.

$$f_x = F / p_x$$

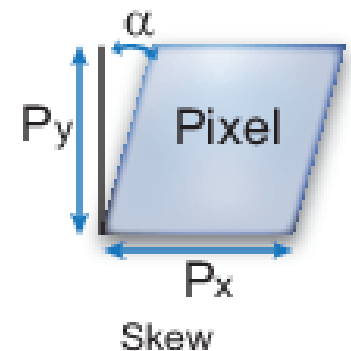
$$f_y = F / p_y$$

F – Focal length in world units, typically expressed in millimeters.

(p_x, p_y) – Size of the pixel in world units.

s – Skew coefficient, which is non-zero if the image axes are not perpendicular.

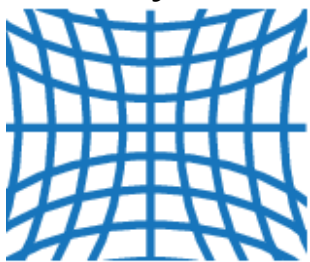
$$s = f_x \tan \alpha$$



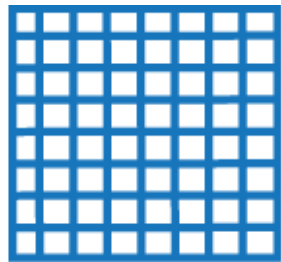
Distortion in Camera Calibration

Lens distortion requires additional correction.

- **Radial Distortion:** Light rays bend more near the edges of a lens than they do at its optical center.



Negative radial distortion
"pincushion"



No distortion



Positive radial distortion
"barrel"

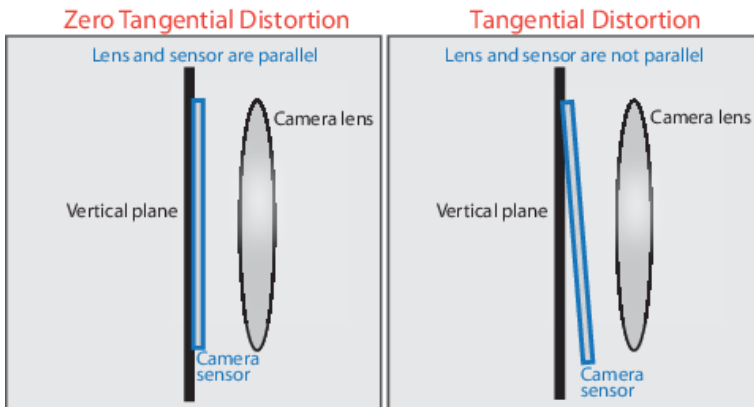
$$r^2 = x^2 + y^2$$

$$x_{\text{distorted}} = x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

$$y_{\text{distorted}} = y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

radial distortion coefficients

- **Tangential Distortion:** when the lens and the image plane are not parallel



$$x_{\text{distorted}} = x + [2 * p_1 * x * y + p_2 * (r^2 + 2 * x^2)]$$

$$y_{\text{distorted}} = y + [p_1 * (r^2 + 2 * y^2) + 2 * p_2 * x * y]$$

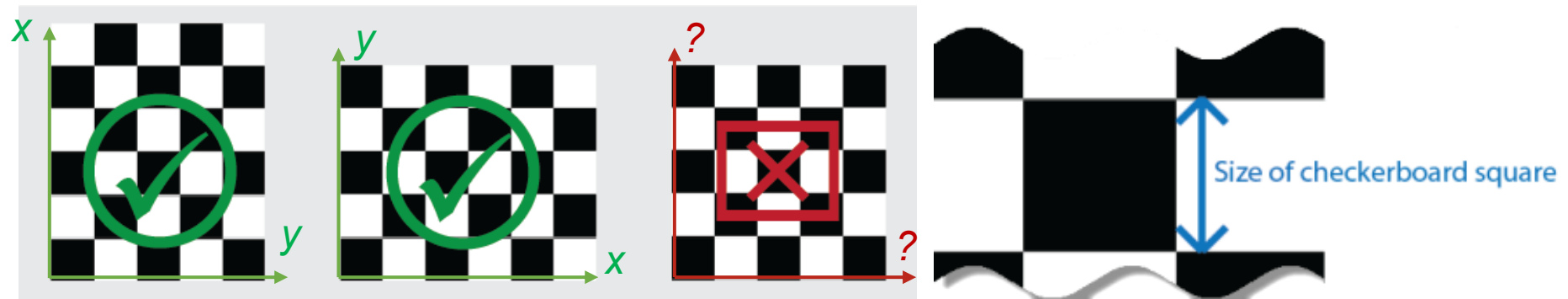
tangential distortion coefficients

Calibration Process

Suppose you already have an algorithm to calibrate the camera.
Can you design a calibration protocol?

Prepare Images

- Print the checkerboard pattern and attach it to a flat surface.
- Measure the size of the checkerboard square for future use.



Even number of squares along one edge and an odd number of squares along the other edge

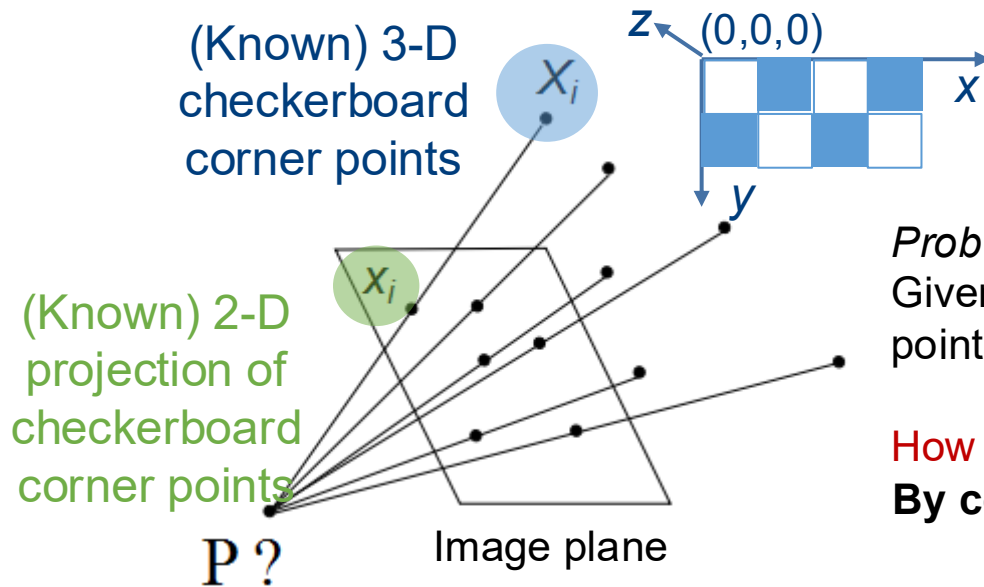
- Capture 10 and 20 images of the calibration pattern. Do not compress the image.



Calibrate

1. Solve for the intrinsics and extrinsics in closed form, **assuming that lens distortion is zero.**

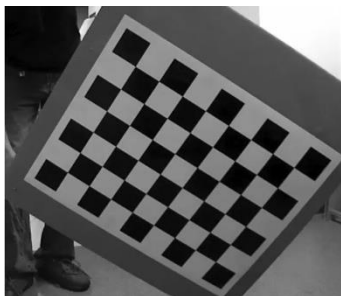
Why not directly solve all parameters including lens distortion?



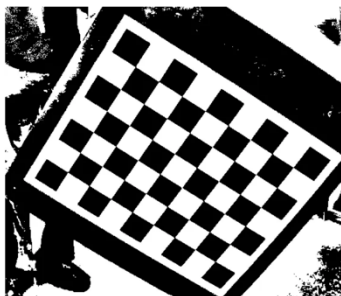
Prob Formulation:

Given the correspondence between 3-D and 2-D points, find the camera matrix P .

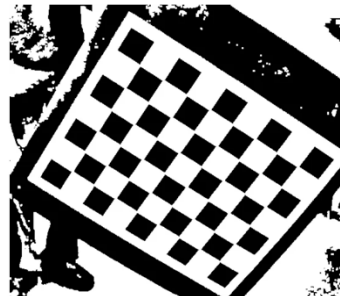
How to find the correspondence?
By corner detection!



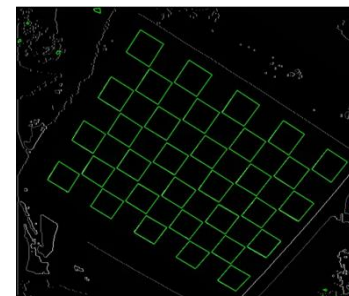
raw image



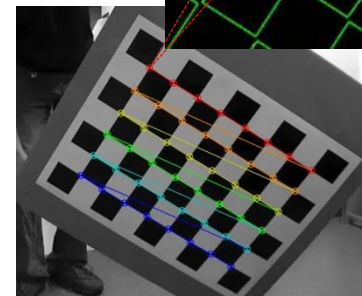
binarization



dilation



quadrilateral detection



result

With the correspondence, we have the **closed-form solution**. Details omitted, please refer to the paper:

Zhang, Zhengyou. "A flexible new technique for camera calibration." *IEEE Transactions on pattern analysis and machine intelligence* 22.11 (2000): 1330-1334.

A Flexible New Technique for Camera Calibration

Zhengyou Zhang

December 2, 1998

(updated on December 14, 1998)

(updated on March 25, 1999)

(updated on Aug. 10, 2002; a typo in Appendix B)

(last updated on Aug. 13, 2008; a typo in Section 3.3)

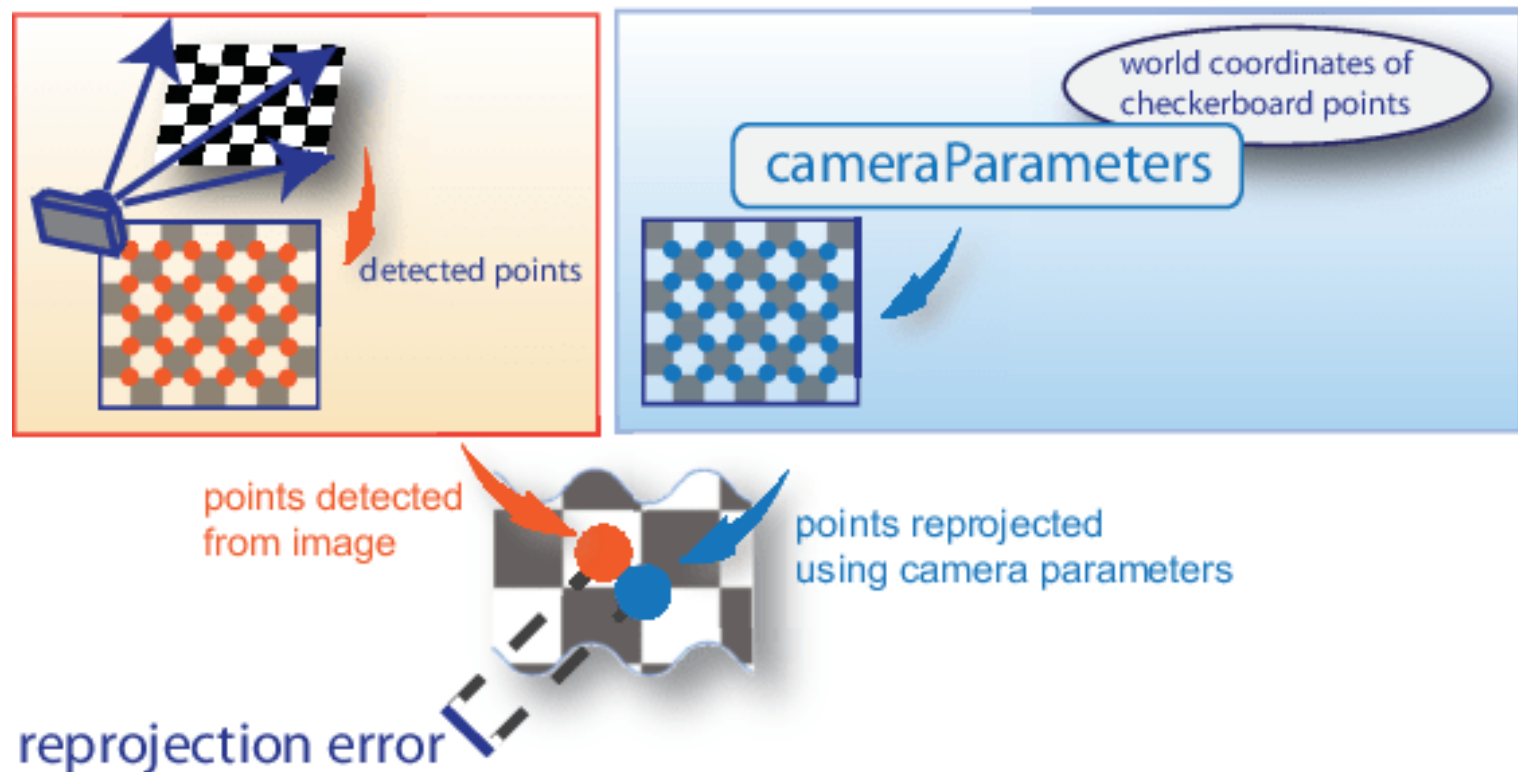
Technical Report

MSR-TR-98-71

Microsoft Research
Microsoft Corporation
One Microsoft Way
Redmond, WA 98052

Calibrate

2. Refine above results by minimizing the reprojection error. Estimate all parameters simultaneously, including the distortion coefficients, using **nonlinear least-squares minimization** (Levenberg–Marquardt algorithm).

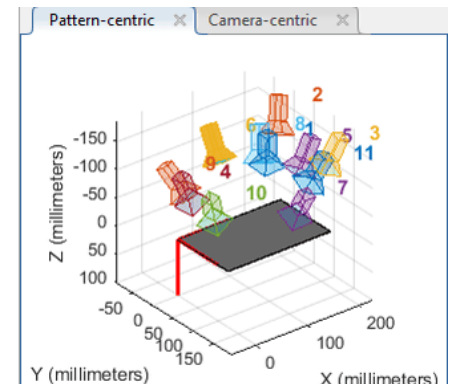
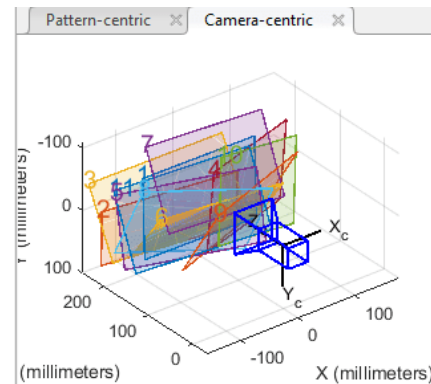


Evaluate Calibration Results

- Check the Reprojection Errors Bar Graph
- Check the Extrinsic Parameter Visualization
- Check undistorted image



Reprojection Errors Bar Graph



Extrinsic Parameter Visualization



Correct distorted-undistorted image pair



Incorrect distorted-undistorted image pair

[Single Camera Calibrator App - MATLAB & Simulink \(mathworks.com\)](#)

Improve Calibration

- Add or Remove Images
 - Add images if there is **<10 images**, the pattern do not cover the whole frame, or the pattern do not have enough variation in orientation w.r.t. the camera.
 - Remove those images with a **high mean reprojection error**, **blurred images**, or images with **incorrectly detected checkerboard points**.
- Change the number of radial distortion coefficients
 - Typically, two coefficients are sufficient for calibration. For severe distortion, such as in wide-angle lenses, you can select 3 coefficients to include k_3 .
- Any other possible ways to improve calibration:
 - Compute the skew parameter
 - Compute tangential distortion

A complex camera model may not be always better!

Event Camera Calibration

- What makes it different for **event camera calibration**?
 - Camera model? Lens distortion?
 - **Event camera output nothing when there is no motion...**
- Ideas
 1. Design a calibration pattern that constantly triggers events, or
 2. Reconstruct events into frames, such that standard calibration patterns can be used.

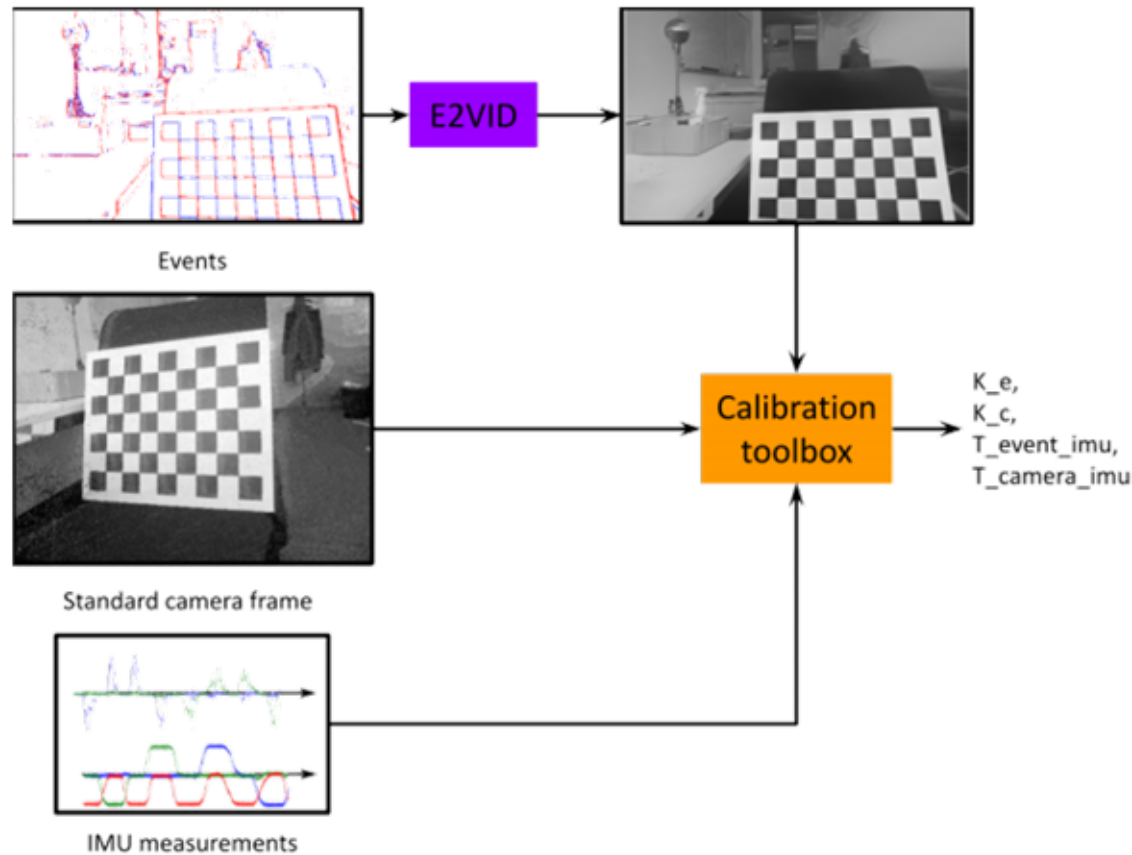
Event-Based Camera Calibration

Hanme Kim

Robot Vision Group
Imperial College London

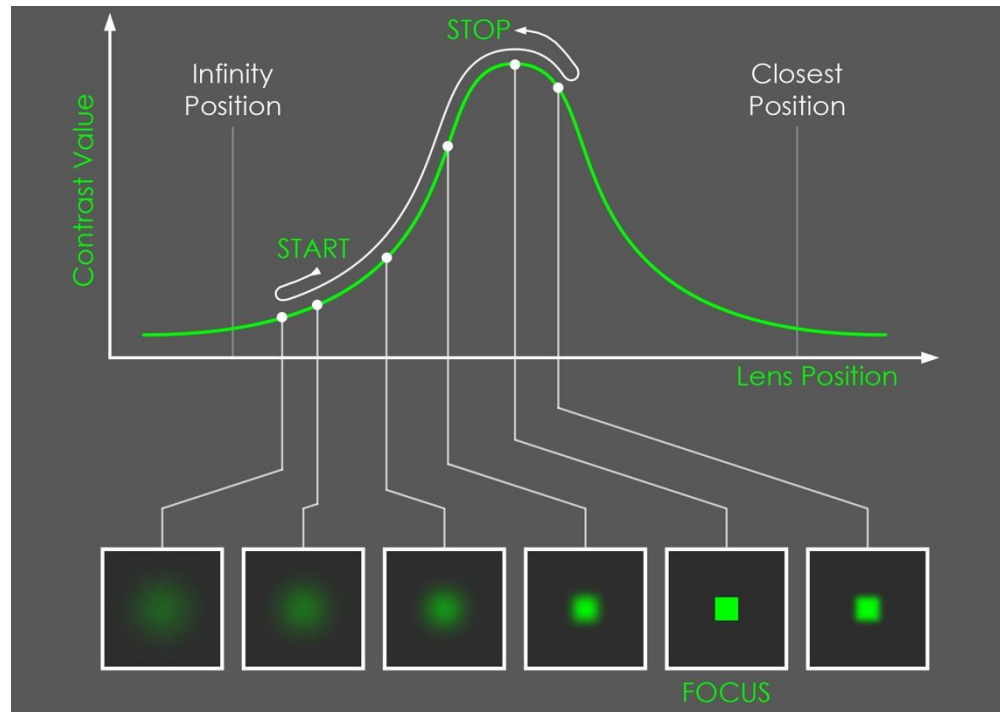
Calibration by Reconstructing Events into Frames

- Intrinsic calibration for event camera
- Multi-sensor extrinsics calibration (conventional camera, event camera, and IMU)



Event Camera Autofocus

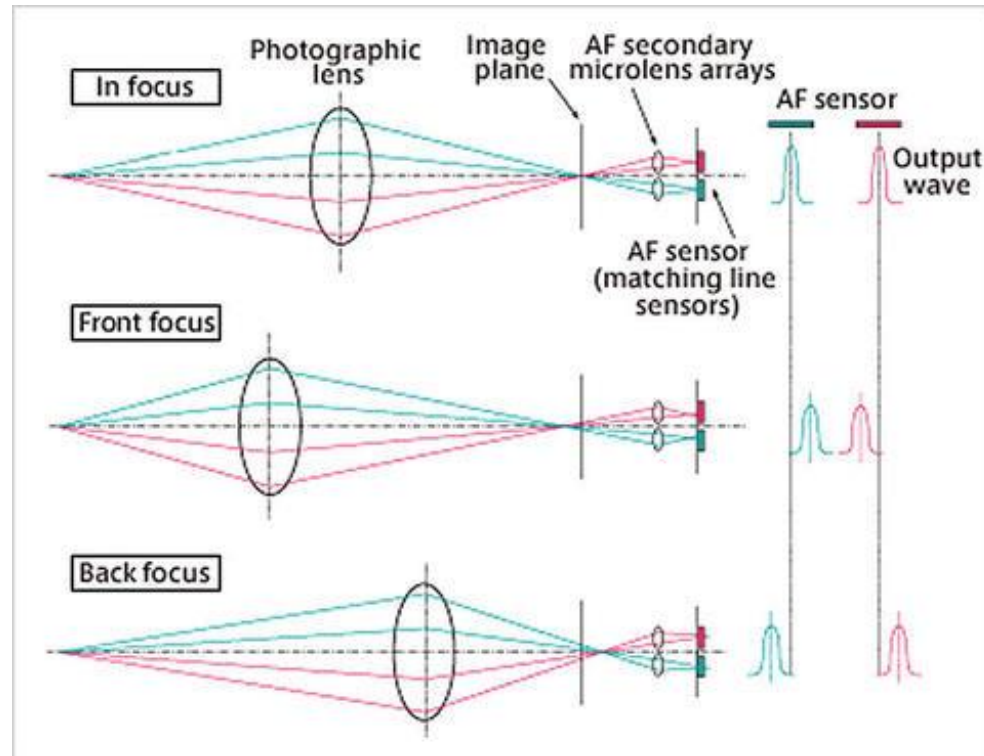
- Autofocus optical system
 - **Active:** by measuring the distance between subject and camera
 - **Passive:** by analysing the measured image
 1. contrast detection



<https://www.photoreview.com.au/tips/shooting/improving-auto-focus-performance/>
<https://www.cambridgeincolour.com/tutorials/camera-auto-focus.htm>
Tech of Digital & Software products (techdigitalsoftwareproduct.blogspot.com)

Event Camera Autofocus

- Autofocus optical system
 - **Active:** by measuring the distance between subject and camera
 - **Passive:** by analysing the measured image
 - 2. phase detection



Fast!

Pros and cons for contrast detection and phase detection for autofocus?

Event Camera Autofocus

- Autofocus for Event Cameras (CVPR 2022)



Autofocus for Event Cameras

Shijie Lin¹, Yinqiang Zhang¹, Lei Yu², Bin Zhou³, Xiaowei Luo⁴, Jia Pan¹



香港大學
THE UNIVERSITY OF HONG KONG



香港城市大學
City University of Hong Kong

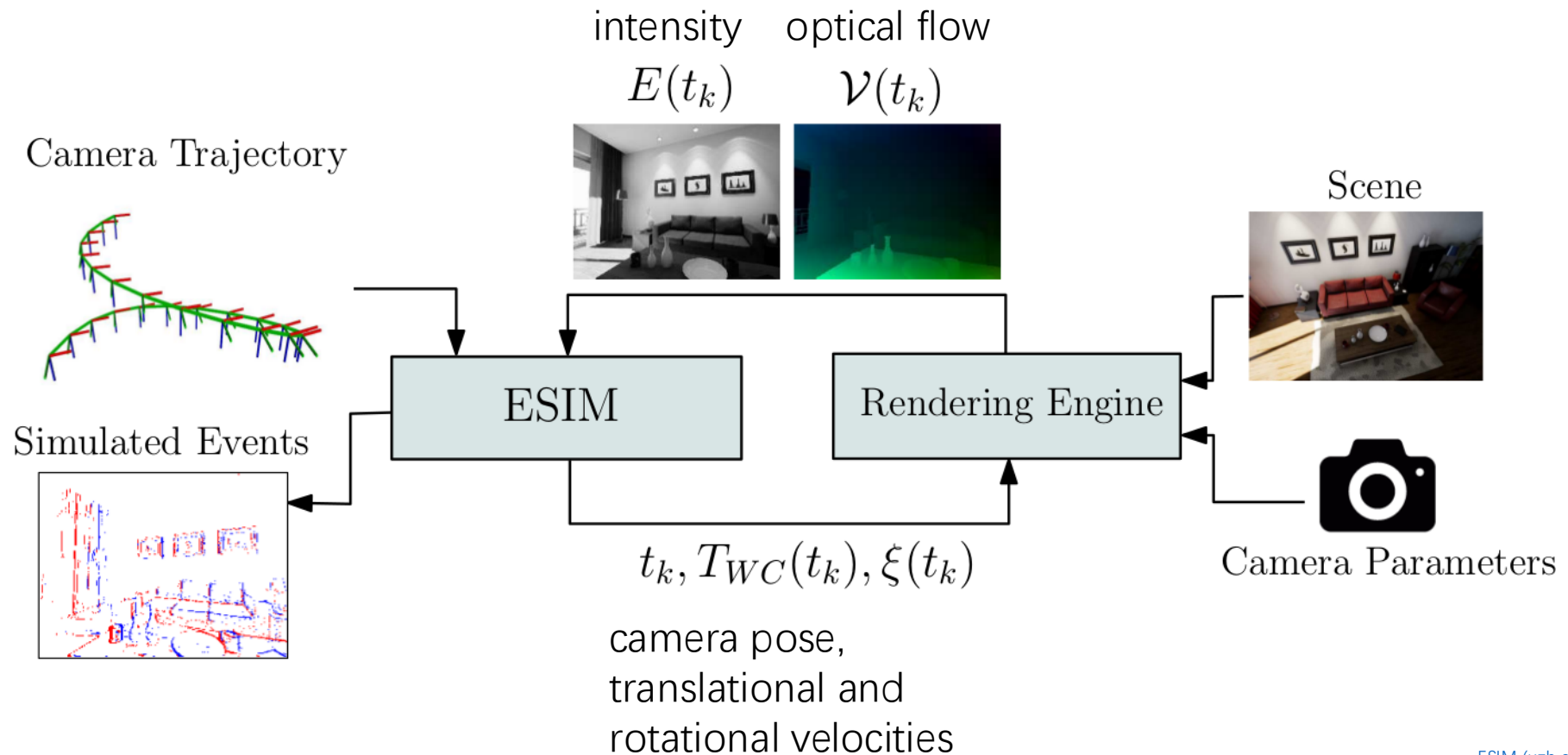
Event Data Simulation

- Why need event data simulation?
 - Not everyone has an event camera!
 - To configure parameters in event generation process
(To freely control the event generation:
noise level, resolution, triggering threshold, ...)
 - To help build the event sensor model mathematically.

Event Data Simulation

- Event data simulators

- [ESIM](#)



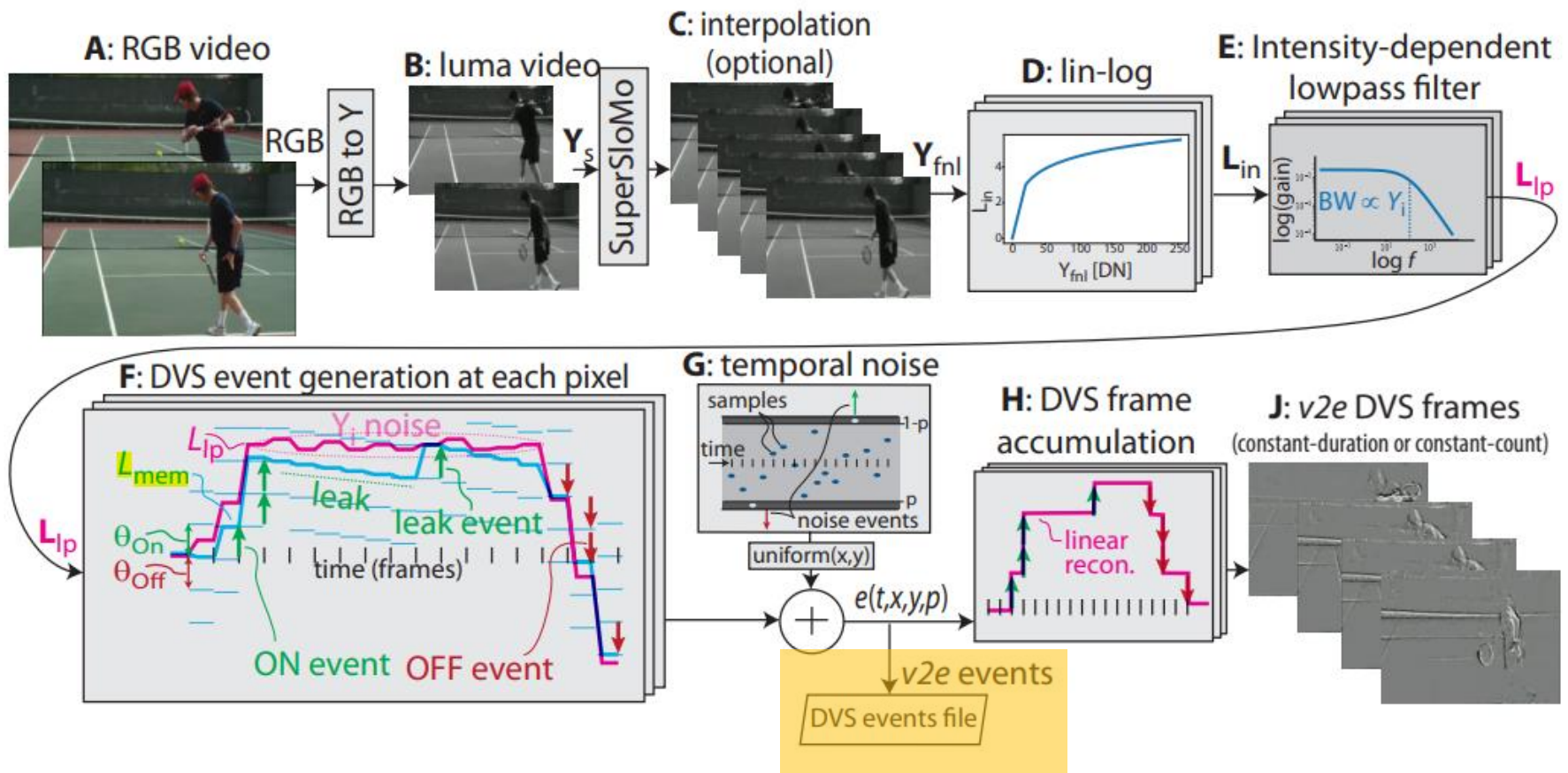
ESIM: an Open Event Camera Simulator

Henri Rebecq, Daniel Gehrig, Davide Scaramuzza



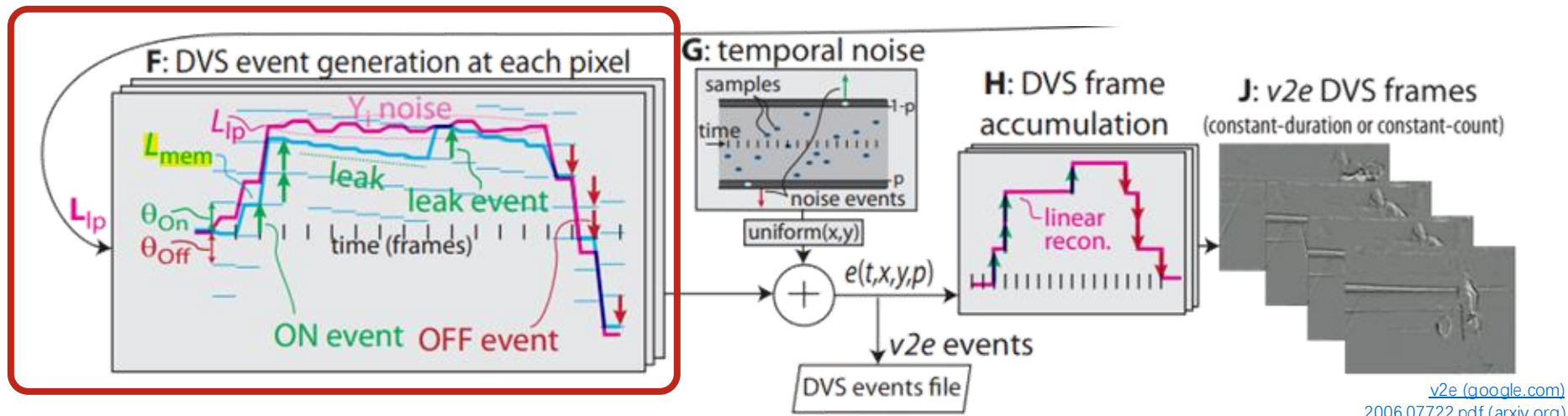
Event Data Simulation

- Event data simulators
 - [v2e](#) (2021, CVPR Workshop on EVS)



Event Data Simulation

- v2e: Noises in event generation model
 - **Threshold mismatch:** Typical value $\theta = 0.3$.
Threshold varies with a Gaussian distribution $\sigma_{\theta} \approx 0.03$.
 - **Hot pixels:** EVS sensors always have some 'hot pixels', which continuously fire events at a high rate even in the absence of input.
 - **Leak noise events:** EVS pixels emit spontaneous ON events called leak events with typical rates ≈ 0.1 Hz.



Event Data Simulation

- v2e vs ESIM

ESIM	v2e
Complex front-end: Can simulate camera motion	Simpler front-end: Only process movies without modelling 3D environments
Programmed in C++	Programmed in Python
Basic pixel model: Contrast thresholds with Gaussian noise for the whole sensor	More realistic pixel model: Pixel to pixel Gaussian temporal contrast threshold variation - Finite, intensity-dependent photoreceptor bandwidth Leak events (intensity-dependent background activity noise) Intensity-dependent temporal noise



- The Multi Vehicle Stereo Event Camera Dataset (MVSEC)
 - Good for **optical flow estimation, pose estimation, and SLAM**

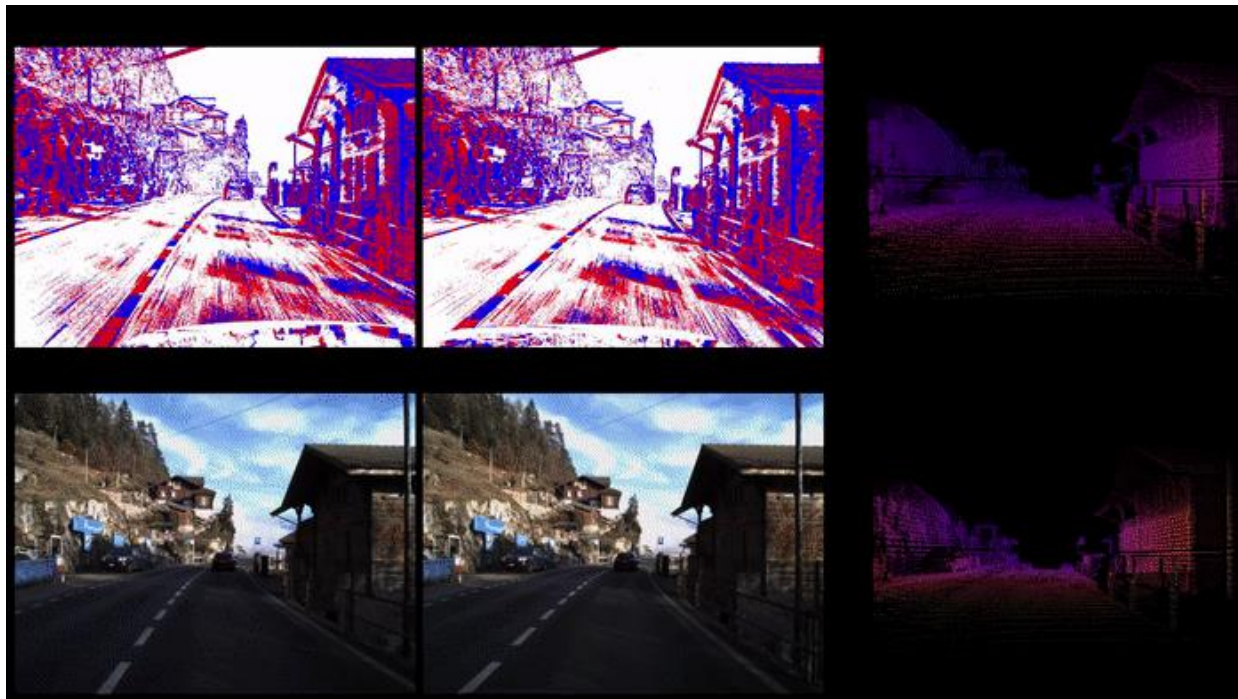
The Multi Vehicle Stereo Event Camera Dataset: An Event Camera Dataset for 3D Perception

Alex Zihao Zhu, Dinesh Thakur, Tolga Özaslan, Bernd Pfrommer,
Vijay Kumar and Kostas Daniilidis

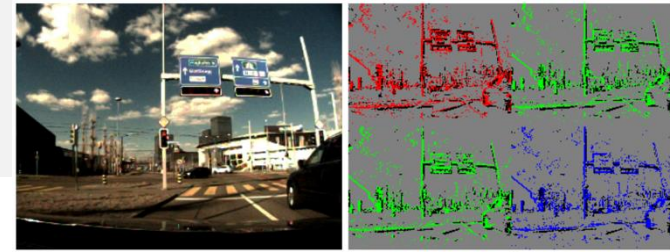


Event Datasets

- A Stereo Event Camera Dataset for Driving Scenarios (DSEC Dataset)
 - **Stereo camera dataset in driving scenarios**
 - 2 monochrome event cameras + 2 global shutter color cameras
 - Lidar data and RTK GPS measurements



Event Datasets



DAVIS frame

Events

- CED: Color Event Camera Dataset
 - Featuring 50 minutes of footage with both color frames and color events from the Color-DAVIS346.

CED: Color Event Camera Dataset

**Cedric Scheerlinck*, Henri Rebecq*, Timo Stoffregen,
Nick Barnes, Robert Mahony, Davide Scaramuzza**

* These authors contributed equally



Australian
National
University



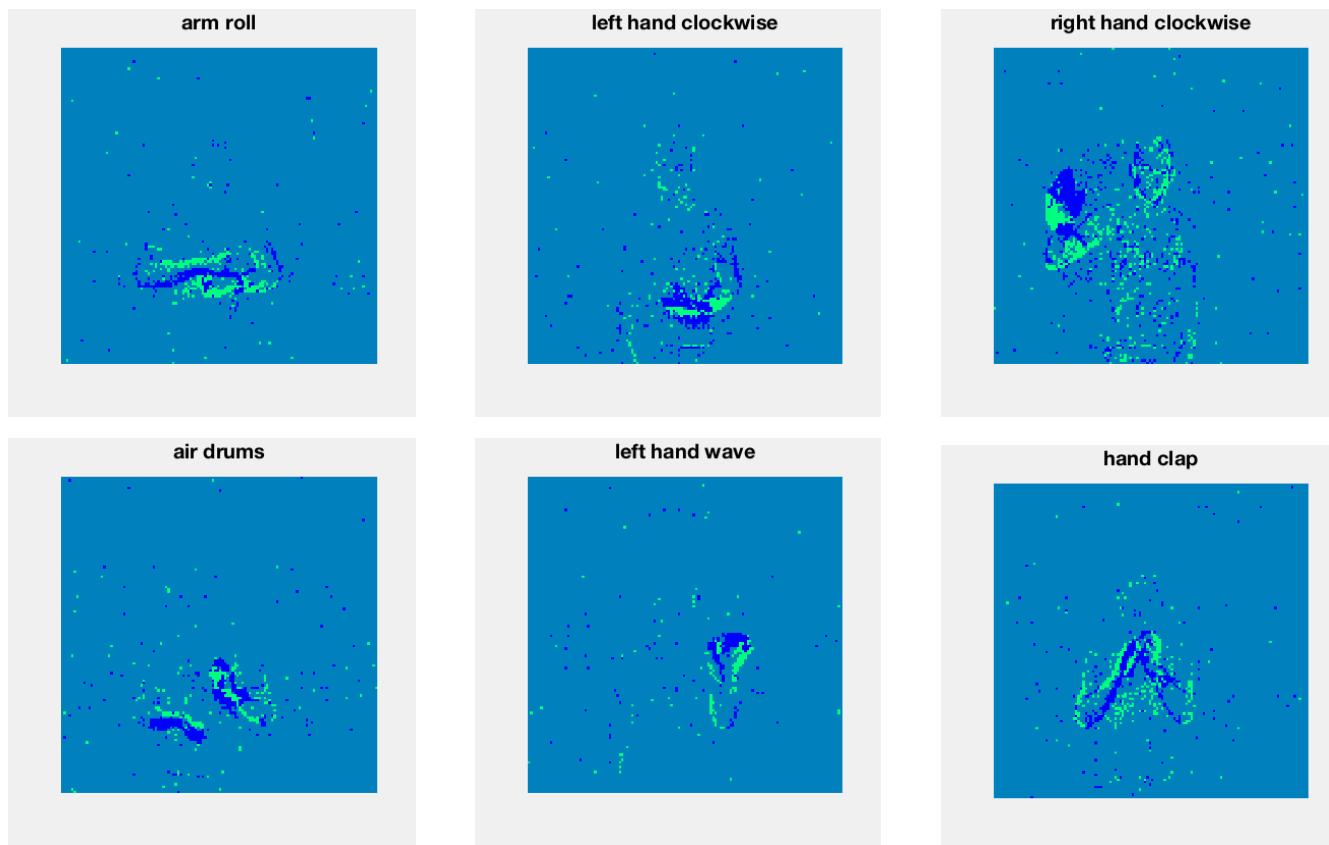
University of
Zurich^{UZH} | **ETH** zürich
Dep. of Informatics – Dep. of Neuroinformatics



MONASH
University

Event Datasets

- DVS128 Gesture Dataset
 - 11 hand gestures from 29 subjects under 3 illumination conditions
 - Can be used to build the **gesture recognition system**

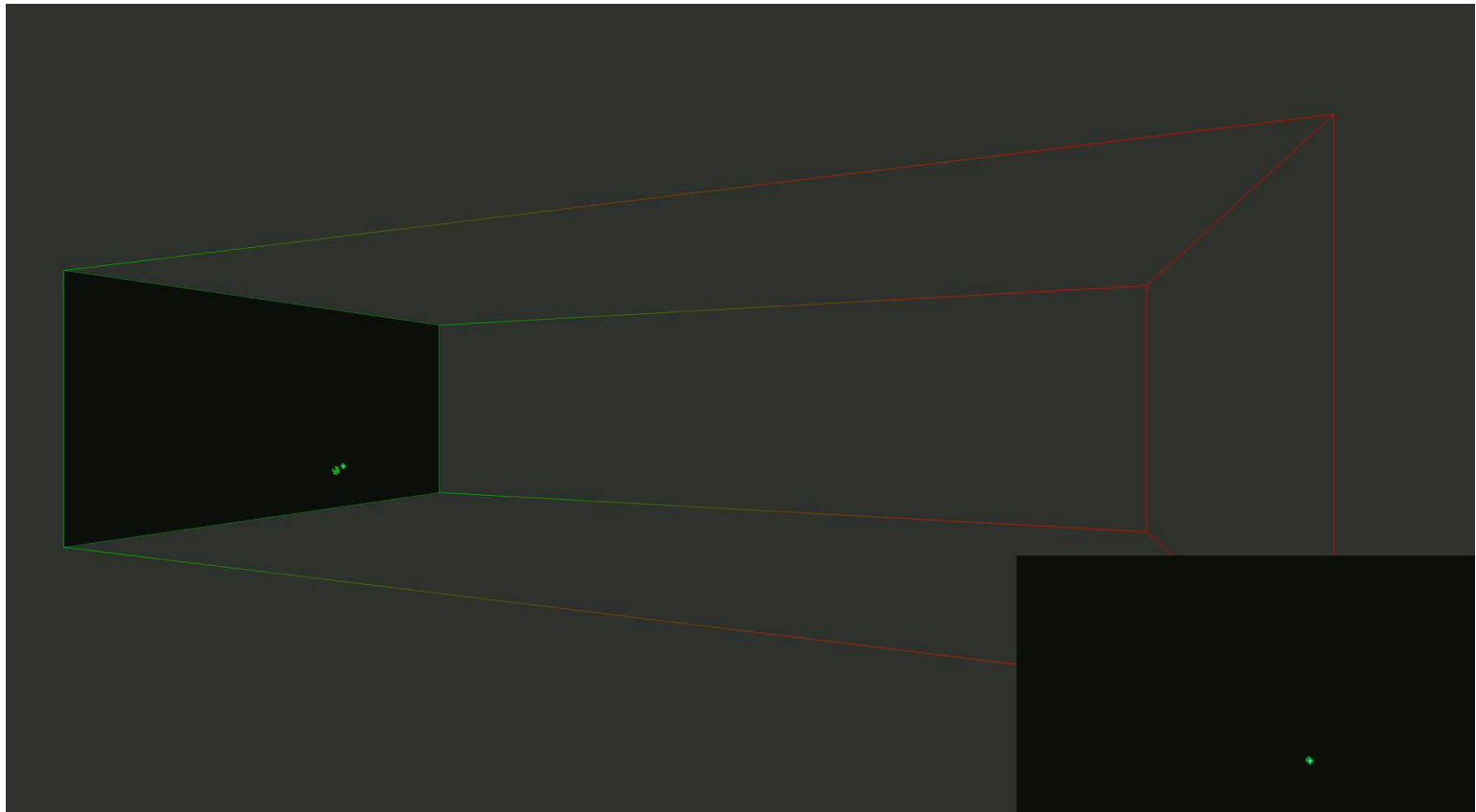


Summary – Event Camera Vision Systems in Real-World Environments

A single EVS chip is not enough in the wild.

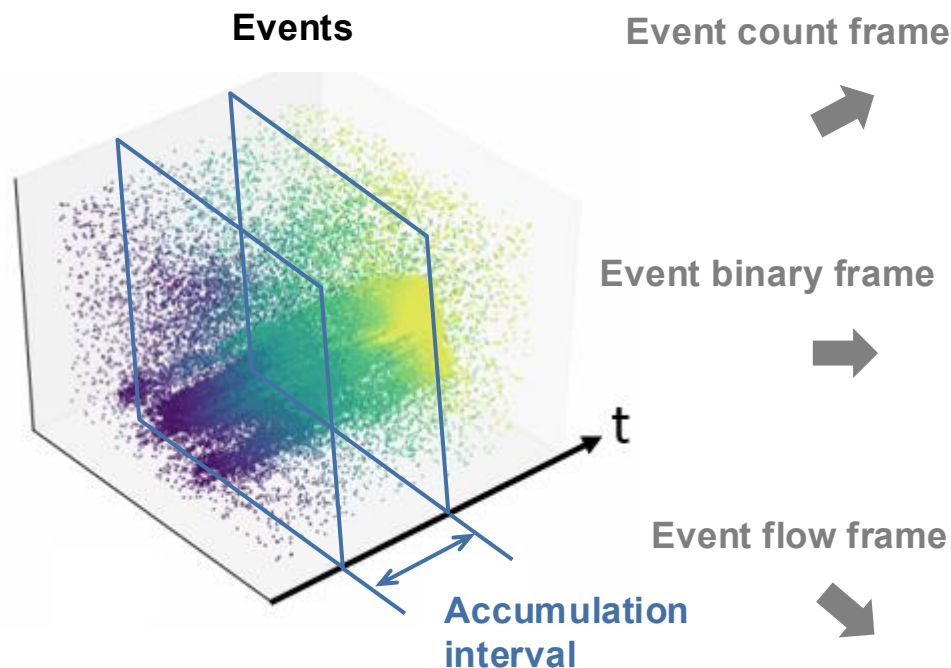
- To enhance data quality:
 - Event camera calibration
 - Event camera autofocus
- To augment EVS database:
 - Event data simulation
 - Event datasets

Event Data Representation: Raw Data (Event Point Cloud)



- + All information retained
- Requires algorithm that can accept point cloud as input

Event Data Representation: Event Frames



4	5	5	6	2
5	4	2	1	2
5	4	6	3	3
0	6	6	5	3
1	0	2	0	0

The number of triggering times

0	1	0	1	0
0	0	0	1	0
0	0	0	0	0
0	1	1	1	0
0	0	0	0	1

If the pixel has been triggered

1	1	1	0	0
2	2	2	0	0
3	3	3	0	0
4	4	4	0	0
5	5	5	0	0

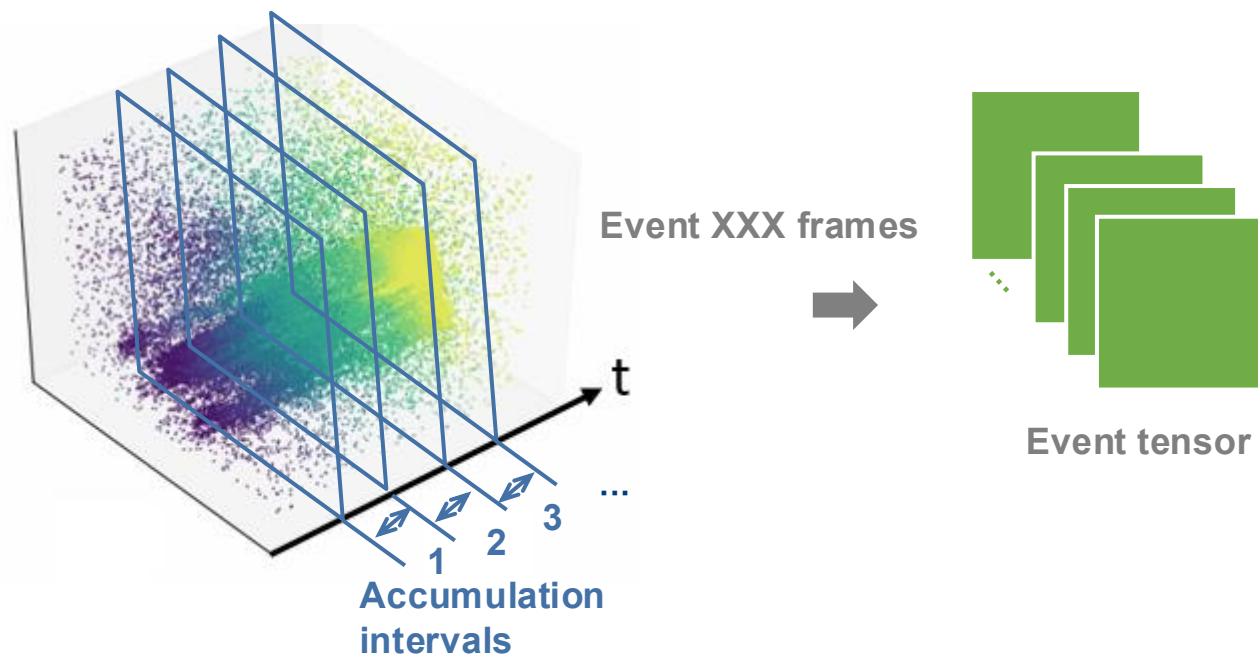
The triggering time of the pixel

Different types of event frames

- + Easy to process by off-the-shelf machine vision algorithms
- Information loss
- Only contains events in accumulation interval

Event Data Representation: Event Tensors

- Adding another dimension to event frames, representing long-time events



- + Easy to process by off-the-shelf machine vision algorithms (especially for deep learning)
- + Represents events in longer time interval without loss too much information
- Requires experiments to determine the number and length of accumulation intervals for best performance, depending on different application

Influence of Event Representation on Data Rate

- **Exercise:** Suppose each pixel triggers **1 time per second** on average; The resolution of the sensor is **1 Mega Pixels**.
Event (x, y, p, t) encoding is (**uint10, uint10, binary, uint32**).
- Calculate and compare the data rate for:
 - event point cloud
 - 30 fps event binary frames (binary)
 - 30 fps event count frames (uint8)

Influence of Event Representation on Data Rate

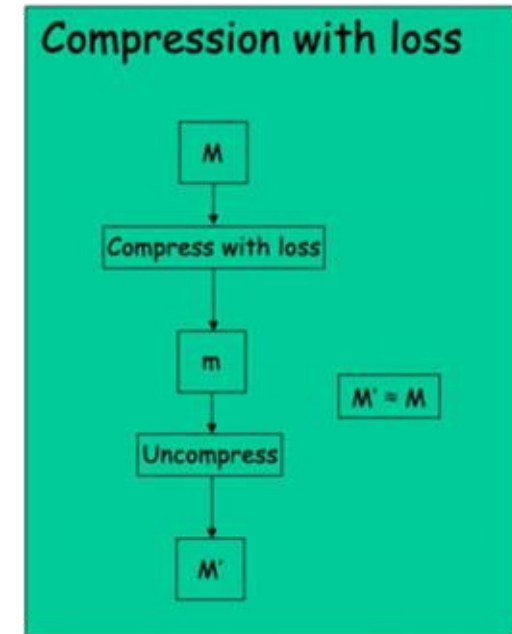
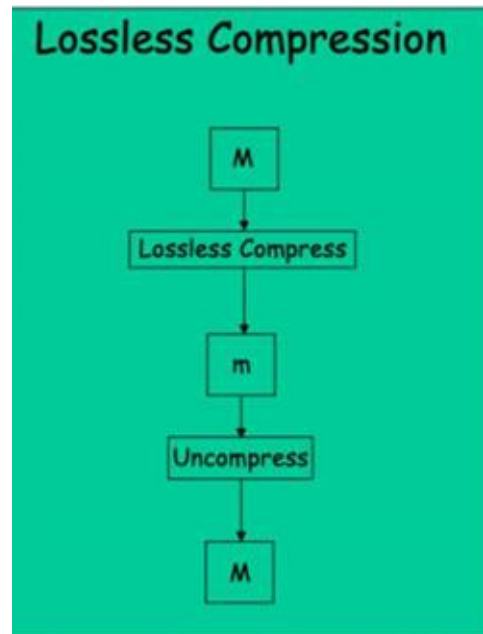
Solution:

Note:

- The data rate for event frames **does not change** w.r.t. event rate, with a **fixed event frame rate**.
 - Data rate more stable. Good or bad? Depending on your algorithm!
- The data rate depends on particular scenarios. E.g., **what if each pixel triggers 10 times per second?**
 - The event frame representation will have much smaller data rate!

Event Data Compression

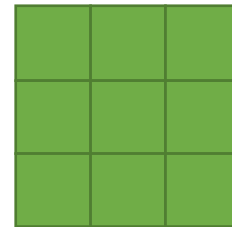
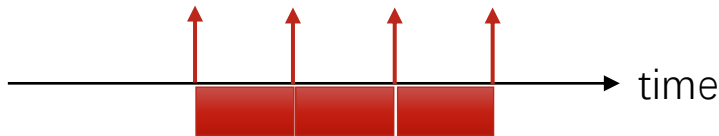
- Some people say, “Event data rate is already much less than conventional camera.” **Why is data compression still required?**
 - Above statement is not always true. In practice, the event data rate varies and depends on the scenario.
 - It is always better to use less bandwidth/space to transmit/store data!
- Two main streams for event data compression:
 - **Lossy compression**
→ Event frames, event tensors are lossy compression.
 - **Lossless compression**
→ The main research direction for event data compression.



Event Data Compression

- Ideas
 1. General data compression approaches can still be applied.
 2. EVS-specific compression approaches. These approaches utilizes: **temporal redundancy**, and **spatial redundancy** of events.
- Temporal redundancy (**left**) and spatial redundancy (**right**)

assuming **events timestamp differences are mostly equal**



assuming **events are triggered in groups within a small neighborhood.**

Reason: adjacent pixels receive almost the same luminance concurrently

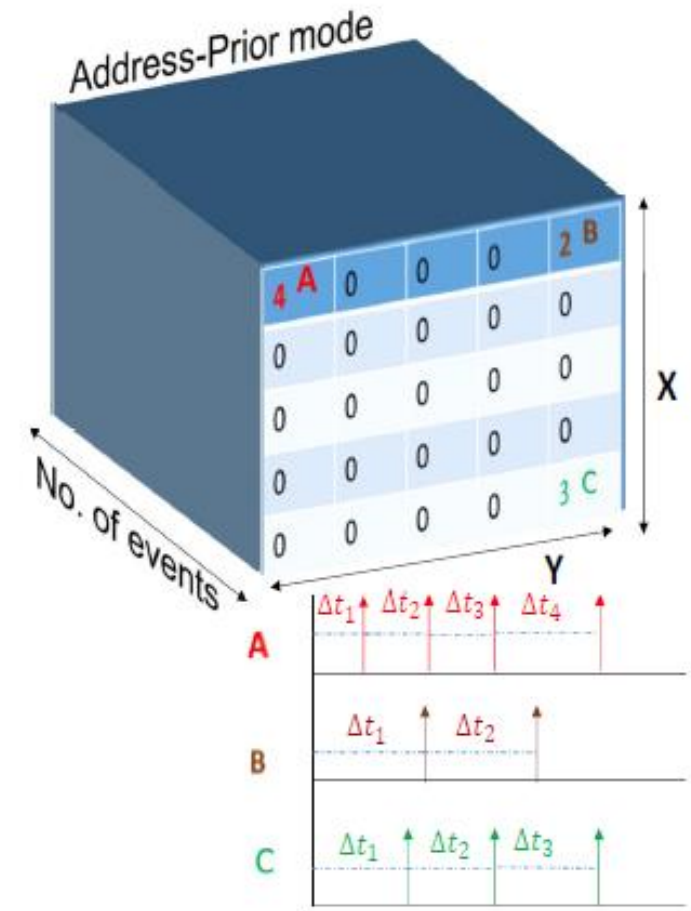
Two modes based on above redundancies are available for event compression.

Address-Prior Mode

- Exploits **temporal redundancy**
 - The number of events at each pixel is recorded, as shown in the fig.
 - We do “**delta coding**” on **time**.

We could use less bits to represent Δt , based on the temporal redundancy.

→ **Efficient compression on timestamp field.**



Time-Prior Mode

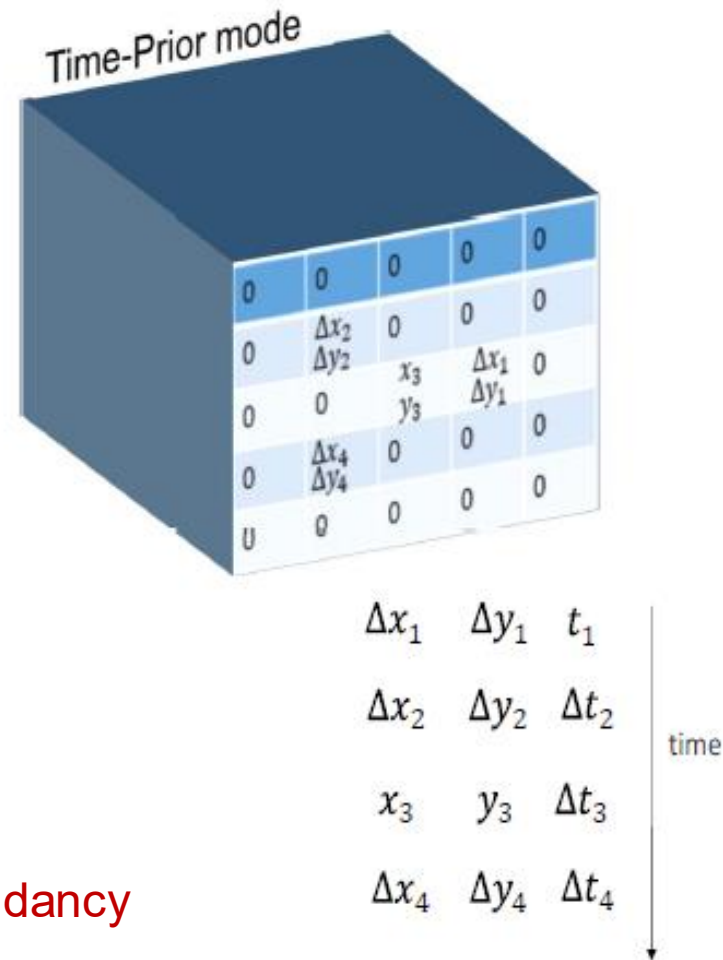
- Exploits **spatial redundancy**
 - Finds a centre point (x_3, y_3 in the fig).
 - Projects all other events w.r.t the centre point.
- We call this “**delta coding**” on **x** and **y**.

We could use less bits to represent Δx and Δy , based on the spatial redundancy.

→ **Efficient compression on address field.**

Drawback: Not suitable for events in a large area.

Open question: can we utilize the temporal redundancy and spatial redundancy simultaneously?



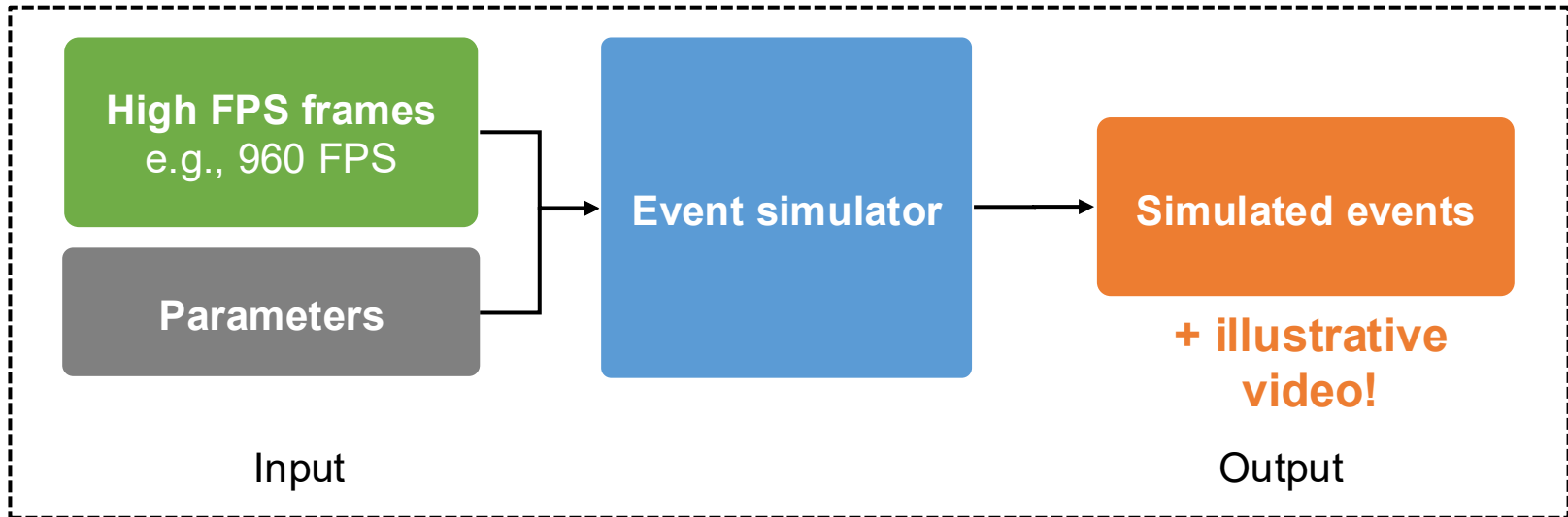
Summary – Event Data Pre-Processing

The raw EVS data may not meet the needs of downstream applications.

- To cater to various needs while reducing the data rate:
 - Event data representations
Raw data – event “point cloud”, Event frames, Event tensors
 - Event data compression
Temporal redundancy and spatial redundancy

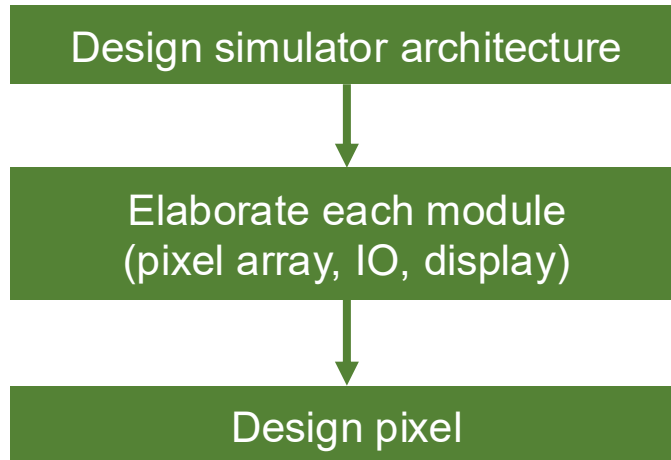
Discussion on CA: Design an Event Data Simulator

- The principle of event camera and event data simulation have been discussed in today's lecture.
- **Aim:** To design a simulation framework to generate events, given high frame-rate images as inputs. A reference diagram is shown below:

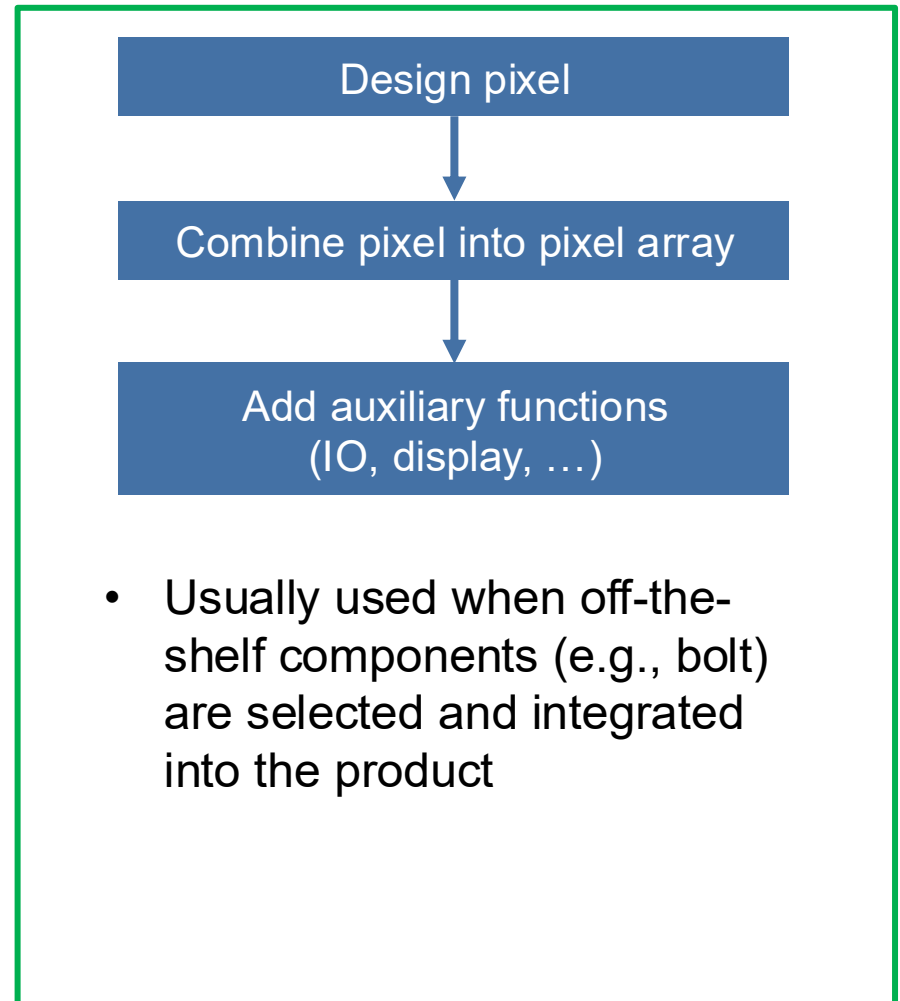


Discussion on CA: Design an Event Data Simulator

- **Top-down** or **bottom-up** design philosophy: Which one would you like?



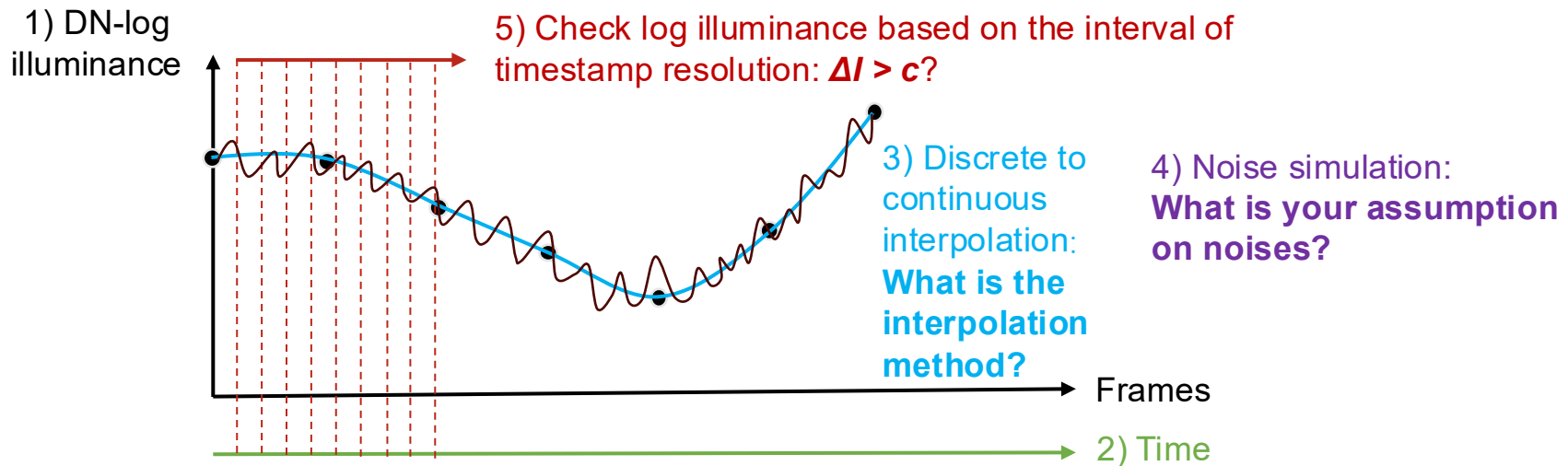
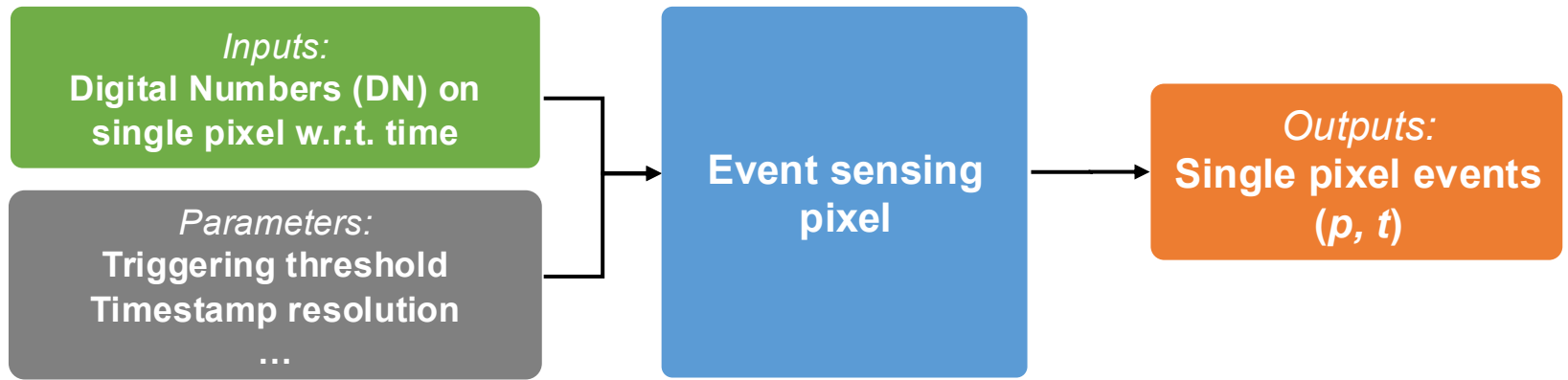
- Usually used for product with more restrictive requirements
- Almost every components is customized
- e.g., a space-suit



- Usually used when off-the-shelf components (e.g., bolt) are selected and integrated into the product

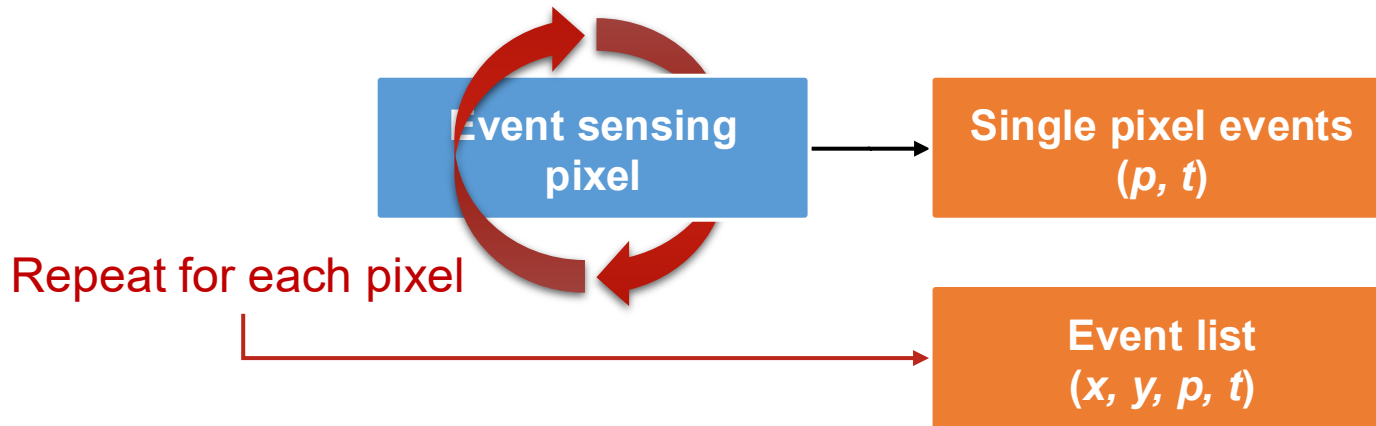
Discussion on CA: Design an Event Data Simulator

- For bottom-up method: **Pixel design**



Discussion on CA: Design an Event Data Simulator

- For bottom-up method: **Pixel array design**



- In practice: a **scanning mechanism/arbitration logic** within the sensor
Could you propose any scanning mechanism?
- If not, a naïve but effective alternative is to take each pixel independently – easy to be implemented within a loop!

Discussion on CA: Design an Event Data Simulator

- For bottom-up method: **IO design**
 - Input: a video clip. Supporting multiple FPS?
 - Output: an event list.
 - Visualization: Event frames on top of the input video.
Supporting customized accumulation time?
- Other considerations:
 - Running efficiency?
 - User-friendly interface / visualization?
 - Output data compression?And more...

References

1. [Event-based Vision: A Survey \(arxiv.org\)](#)
2. [CIS 849 -- Autonomous Robot Vision \(udel.edu\)](#)
3. [Course Detail \(imaging.org\)](#)